

Subject-Bound Evidence Composition for Delegated AI Actions

Executable Indistinguishability Witnesses and Commit-Time Revocation

ANTON SOKOLOV, Tyche Institute, Estonia

Background: Delegated AI workflows can contain individually valid policy, evidence, runtime-state, authority, measurement, and effect artifacts and still provide no end-to-end basis for the one action they are meant to authorize.

Objectives: We show constructively that this failure class is representable, localizable, and repairable at the composition boundary, by evaluating a typed, non-compensable decision rule over five artifact predicates plus an explicit cross-artifact binding stage over the implemented effect, resource, report-time, and measurement-profile coordinates, and by testing commit-time revocation and typed first-failure transfer under stronger evidence classes.

Methods: Three executable laboratories cover signed policy-version/evidence replay, SAFE-metric metamorphics, and a frozen 104-transaction composed corpus with runtime-state, delegation, and binding checks. Companion laboratories add Ed25519 authority credentials, generated single- and multi-fault mutations, scheduled and live revocation, 104 transaction-bound TPM2 quotes under eight software-TPM roots plus an identified Hyper-V VM exposing a Microsoft vTPM, a separately compiled native verifier, a finite Java model check, process-isolated and five-container durable revocation, and typed transfer of a public two-language artifact-verifier result. The sealed capsule is executed twice on hosted x86_64 and ARM64 job VMs.

Results: The corpus contains eight denials in which all five artifact verifiers pass and the subject binding fails, each paired with a matched all-valid lookalike that is allowed. Point-only, four-of-five majority, and artifact-validity ablations produce 2, 31, and 48 false allows; the majority count decomposes exactly into 23 single-fault denials plus the 8 cross-layer denials, and partial bindings that omit time or profile leave 1–3. Native appraisal reproduces all 104 state classes and rejects 64/64 predeclared mutations in both implementations; the identified VM verifies 104/104 live quotes and rejects 64/64 mutations. The atomic revocation-status/effect guard has zero safety errors and zero duplicate effects across 372 durable cases with 96/96 fault recoveries, while every read-based profile exposes stored counterexamples. Typed transfer preserves 21/21 native-code rows; a separate SQL path matches 125/125 rows and detects all 46 mutation and 46 omission changes. Both hosted repetitions satisfy 20/20 semantic and 11/11 cross-architecture assertions.

Conclusions: Local Boolean validity cannot substitute for typed composition over the implemented subject coordinates, and revocation status and effect require one commit boundary. All counts are coverage on author-designed synthetic fixtures; the result is an executable benchmark and systems contribution, not a deployed-system rate, physical-host attestation, hardware-root, human-agreement, or interoperability claim.

1 Introduction

An agent may present a valid identity and execute a schema-valid tool call. Its machine may report an affirming attestation result. The action receipt may be correctly signed and internally well formed while reporting a claimed effect. A model-quality score may exceed a policy threshold. These statements do not entail that the specific effect was both authorized and supported by an applicable measurement.

The gap is compositional, and it is a classical one: component certificates do not compose into a system certificate without explicit discharge of the assumptions each certificate rests on (Rushby 2002). Each verifier here answers a scoped question:

- a policy verifier identifies which appraisal rule was active;
- an evidence verifier checks integrity and freshness;
- a state verifier consumes a typed appraisal result; the frozen source fixtures encode it structurally, and the native overlay re-derives it from transaction-bound TPM2 evidence;
- an authority verifier checks whether a delegation path covers an action, with a terminal effect check against the receipt inside its own case file;
- a measurement verifier checks profile applicability and inferential support.

Author’s Contact Information: Anton Sokolov, Tyche Institute, Tallinn, Estonia, ORCID: [0000-0003-2452-7096](https://orcid.org/0000-0003-2452-7096), anton.sokolov@tyche.institute.

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Effect is not a sixth artifact with a sixth verifier. It is a subject that several artifacts describe at once, and no single verifier can see all the descriptions. The policy artifact carries the required effect and the required measurement profile; the evidence artifact carries the effect, resource, and issue time actually reported; the authority artifact carries the terminal narrowed scope and the interval in which the whole delegation path is valid; the measurement artifact carries the profile the appraisal was actually run under. A predicate over all four has no local home. The component interfaces cannot force a mismatch between the canonical action and the receipt's reported effect to surface as an authority or evidence failure: neither layer can see both sides of the comparison.

Local validity may coexist with global denial. For example, a signed but superseded policy can be authentic; a replayed receipt can retain a valid signature; a point estimate can exceed a threshold while its lower confidence bound does not; and an evidence receipt can validly report a WRITE effect when the valid delegation path permits only READ.

That last case is the worked example, evaluated in both directions. Under a composition of the five scalar layer verdicts alone, the verifier **allows** it: a correctly Ed25519-signed evidence object with effect `ledger.write` on resource `invoice-999` and an unseen nonce, evaluated under a policy whose required effect is `ledger.read` against an ALLOW delegation path whose terminal grant carries only `ledger.read` on `invoice-123`, returned policy PASS, evidence PASS, state PASS, authority ALLOW, measurement PASS, verdict ALLOW, and first rejecting gate verified. With the explicit binding stage, the same transaction returns binding result `EFFECT_MISMATCH`, verdict DENY, and first rejecting gate `binding.effect` — with all five component verifiers still passing, and with a matched lookalike that differs only in the reported subject fields still allowed at gate verified. A standalone acceptance test builds both transactions from first principles rather than reading the frozen corpus, so the result cannot be satisfied by a corpus entry carrying a convenient label (Sections 4, 5.4, and 6.3).

This article evaluates the following decision rule:

$$\text{Allow}(x) = P(x) \wedge E(x) \wedge S(x) \wedge A(x) \wedge M(x) \wedge B(x),$$

where P is exact policy applicability, E is evidence integrity/freshness, S is runtime state, A is authority-to-effect validity, M is measurement support, and B is the implemented partial binding. The rule is non-compensable: five passing predicates cannot offset one failed predicate. B is not a sixth artifact verifier and introduces no sixth artifact. It is a well-formedness condition on the conjunction over effect, resource, report time, and measurement profile fields extracted from the policy, evidence, authority, and measurement artifacts. The current B does not bind a state/workload subject or the other omitted coordinates listed in Section 4.1. The conjunction is commutative; the *evaluation* order is fixed separately and places B last (Section 4).

The contribution is deliberately artifact-first. Theorem 1 frames the problem as an information limitation and is proved in one line; the advance this article claims is the executable instantiation — a typed decision-transaction model, a frozen counterexample corpus with matched lookalike controls, ablation identities, native state evidence, an atomic revocation boundary, and cross-implementation, cross-architecture replays — for the runtime accountability layer that delegated-agent deployments currently lack. In AI terms, the object is the seam that norm monitoring (Criado 2018) and multi-agent plan diagnosis (Micalizio and Torasso 2014) study at the level of norms and plans, approached here at the level of the evidence one delegated action carries.

The work has three objectives:

1. establish the semantic limit of local validity and make its remedy executable, using explicit policy and measurement semantics, re-executed typed layer evaluators, a cross-layer binding stage, and matched indistinguishability witnesses;
2. test the two runtime boundaries abstracted by the main corpus: native appraisal of a frozen attestation vector and atomic revocation-status checking with effect commit; and

3. test whether the typed first-decisive-state representation transfers to a separately built artifact verifier while preserving its native diagnostic codes.

The result is an executable artifact study, not a claim of production readiness or population-level safety improvement.

2 Research Questions and Claim Boundaries

2.1 RQ1: When is local validity insufficient?

Can individually passing policy, evidence, state, authority, and measurement verifiers be locally indistinguishable between a transaction matched on the implemented coordinates and one mismatched on a required coordinate that must be denied? Within this question we test exact policy state, SAFE measurement semantics, typed composition, partial-binding ablations, named-gate localization, and the number of constructed all-local-pass denials in the frozen corpus.

2.2 RQ2: Where must runtime authority become atomic?

Does native offline appraisal preserve all 104 transaction state classes while rejecting targeted quote, PCR, freshness, and subject-binding mutations, and, in process-isolated signed-status races with durable retry and restart, does a linearizable status-and-effect guard avoid the false allows and duplicate effects exposed by incomplete protocols?

2.3 RQ3: What survives transfer across verifier ecosystems?

Can a separately constructed two-language artifact-verification experiment be represented in the same first-decisive-state form without discarding native rejection codes, and how much diagnostic distinction would a Boolean valid/invalid projection erase?

2.4 Threat model and adversary assumptions

The laboratories model an adversary who can present, substitute, replay, tamper with, or withhold individual artifacts: policies (missing, stale, substituted, re-signed after tampering), evidence objects (tampered payloads, replayed nonces, mismatched subject fields), attestation-result objects and TPM2 quote material (mutated messages, signatures, Platform Configuration Register (PCR) blobs, challenges, bindings), delegation paths (broken lineage, expired edges, amplified scope), measurement fixtures (substituted profiles), and revocation timing (races between status reads, revocation, and effect commit). The adversary cannot break Ed25519 signatures, corrupt the shared composition-rule module, or alter frozen corpora without changing recorded hashes. Key compromise, side channels, denial of service, and supply-chain attacks on the toolchain are out of scope. The corpus instantiates this adversary by construction; no claim is made about attacker prevalence in deployed systems.

2.5 Claim ceiling

The study may report exact observations from frozen artifacts. It does not claim:

- prevalence in deployed agent systems;
- a defect in Veraison, A2A, MCP, RATS, SCITT, SAFE AI, or TOPSIS;
- ownership of the SAFE integration functions of Giudici and Kolesnikov;
- invention of authenticated or transitive delegation;
- externally independent labels;
- production interoperability or legal compliance;
- sufficiency of the proposed passport fields for every environment.

Three additional boundaries apply:

- external independence of the expected labels: all labels are author-written within this research programme, and the composition-level labels are generated by the same shared composition-rule module the composed verifier imports, so agreement at the composition level is analytic (Section 6.3);
- cryptographic attestation in the **source corpus**: its original state predicate is a structural appraisal of a corpus-supplied attestation-result object, with no TPM, no Veraison, and no verification of any runtime. A separate 104-transaction companion supplies transaction-bound TPM2 evidence for every state decision and recomposes the complete tuples, but remains software-TPM evidence on one host rather than attestation of a deployed workload;
- real-world false-allow rates: every false-allow count in this article is a count on a designed corpus, a corpus-coverage statistic, and must not be read as an estimate of prevalence anywhere.

A separate boundary applies to the binding stage. It is a deterministic string-agreement and interval-containment condition over subject fields read out of corpus-supplied artifacts. It verifies no signature, takes no measurement, contacts no attestation service, and does not upgrade any corpus-supplied flag into a cryptographic fact. The denials it produces are denials on a designed corpus, and the claim it supports is representability and localization, not detection performance.

The companion experiments introduce three further ceilings:

- the seven-case native companion is one frozen swtpm vector; the 104-transaction companion uses eight fresh software-TPM roots, but neither experiment supplies a hardware root, live attested workload, manufacturer endorsement, or independent-host evidence;
- the 276-case service is one-process loopback; the durable service separates status, effect, and contender processes with durable linearization and recovery, and a five-container extension adds two interchangeable effect instances plus a deterministic fault proxy. Both remain on one physical host with centralized SQLite linearization rather than multi-host, Internet-scale, consensus, or performance evidence;
- cross-ecosystem transfer means that an explicit analytic crosswalk preserves native first-failure codes from two executable corpora; it does not mean semantic equivalence, protocol interoperability, or independent validation.

3 Related Work and the Remaining Composition Seam

3.1 Measurement integration

Kolesnikov’s master’s thesis proposes an integrated S.A.F.E.-AI compliance score built from accuracy, explainability, and robustness vectors, denoted RGA, RGE, and RGR, and studies arithmetic, geometric, root-mean-square, and TOPSIS integration (Kolesnikov 2025). Giudici and Kolesnikov develop this integrated SAFE measurement line in a journal article and working paper (Giudici and Kolesnikov 2025, 2026). Babaei and Giudici provide a statistical SAFE AI package (Babaei and Giudici 2025).

We do not propose a replacement metric. We test whether an evidence identity distinguishes aggregation, profile, reference-set, dependence, and confidence semantics that a scalar score alone omits.

3.2 Delegation and authorization

South et al. argue that AI agents require authenticated and auditable delegation, including agent-specific credentials and robust scope translation (South et al. 2025). WAVE already provides signed authorization graphs, transitive scoped delegation, revocation, and independently verifiable authorization proofs (Andersen et al. 2019). JEDI provides restricted key delegation across scope and time (Kumar et al. 2019). We note that these two systems share several authors from one research group, so they are related lines of work rather than two independent bounds on novelty; either alone suffices to bound any claim about delegation primitives.

A second adjacent family binds tokens and assertions to their intended subject or audience inside one credential system. The confused-deputy problem (Hardy 1988) names the underlying failure; OAuth 2.0 mitigates it with resource indicators (Campbell et al. 2020), rich authorization requests (Lodderstedt et al. 2023), and sender-constrained tokens (Fett et al. 2023), and SAML audience restriction, XACML condition machinery (OASIS 2013), W3C Verifiable Credentials (World Wide Web Consortium 2025), and SPIFFE workload identities (SPIFFE Project 2026) each bind an assertion to a scope, condition, or workload. General-purpose policy engines such as Open Policy Agent evaluate one policy over an assembled request context (Open Policy Agent 2026). These mechanisms constrain one artifact or centralize one decision; none of them composes five independently verified artifact families while preserving each native typed verdict and a first-decisive-gate localization. A single policy engine evaluating the full request context could encode the binding predicate B — but it would then *be* the composer this article studies, and it would still need the typed inputs, the subject projections, and the witness corpus evaluated here. The contribution is the typed interface and the executable evidence, not a claim that no engine could host the rule.

The remaining experimental seam is the join from a valid authorization path to the canonical action and effect described by evidence. Our authority layer asks whether the terminal action remains entailed after delegation, freshness, status, scope, and binding checks — but it asks that question only of the objects inside its own case file. The join across artifacts belongs to the composition and is implemented there (Sections 4 and 5.4).

3.3 Policy versioning and change impact

The policy-state subquestion of RQ1 has prior art in policy analysis. Margrave verifies access-control policies and computes the semantic impact of policy changes (Fisler et al. 2005), and XACML 3.0 carries an explicit Version attribute on policies and policy sets (OASIS 2013). Those tools answer what a policy means and how an edit changes it. Our subquestion asks a narrower operational question: whether the decision procedure binds each decision to the exact policy identity, version, validity window, and digest that were active at decision time, so that an authentic but superseded or substituted policy cannot silently govern an appraisal. Versioning metadata existing in a policy language does not by itself make the verifier check it; the replay corpus tests the checking.

3.4 Agent benchmarks

TeamBench evaluates coordination under operating-system-enforced role separation (Kim et al. 2026). AgentRedBench includes underspecified-authorization attacks over SaaS integrations (Dingeto and Leeney 2026). ToolPrivacyBench evaluates field-to-tool authorization for private information (Hu et al. 2026). TraceSafe localizes mutated safety failures in tool trajectories (Y.-S. Chen et al. 2026). Heartbeat-Bound Hierarchical Credentials evaluates bounded hierarchical revocation (Deochake 2026). Li benchmarks authenticated account-level actions that leave the user’s assigned task, in a high-performance-computing setting (Li 2026).

These works preclude broad claims such as the first unauthorized-action, role-isolation, trajectory-localization, or cascading-revocation benchmark. Our narrower object is typed composition: separately appraised artifacts that become invalid only when exact policy, evidence, state, delegation, measurement, and effect relations are joined.

3.5 Attestation and transparent evidence

Remote ATtestation procedures (RATS) separates Evidence, Verifier appraisal policy, reference values, and Attestation Results (Birkholz et al. 2023). Supply Chain Integrity, Transparency, and Trust (SCITT) distinguishes signed statements, transparency registration, receipts, and later audit (Birkholz et al. 2026). A2A and MCP define agent interaction and tool-facing authorization contexts (Linux Foundation 2026; Model Context Protocol 2025).

NIST’s software and AI agent identity initiative identifies authorization, auditing, and non-repudiation as active infrastructure problems (NIST NCCoE 2026).

These mechanisms motivate typed boundaries. An attestation result is not an action grant; a transparency receipt is not proof of measurement adequacy; and an authenticated tool call is not necessarily within a delegated task. The composition question, what discharges the assumptions between such certificates, is the modular-certification problem in an agent setting (Rushby 2002).

3.6 Closest JAIR neighbours: artifacts, diagnosis, and norms

Belardinelli, Lomuscio, and Patrizi formalize artifact-centric multi-agent systems with data/lifecycle semantics and temporal-epistemic model checking (Belardinelli et al. 2014). Micalizio and Torasso diagnose expected-effect failures in distributed, partially observable multi-agent plans, including dependent failures (Micalizio and Torasso 2014). Criado treats norm monitoring as a resource-allocation problem under incomplete observations and proves properties of a resource-bounded monitor (Criado 2018). These are closer technical neighbours than credential syntax.

Our narrower object is a single authorization transaction at one decision boundary. We neither model-check an artifact lifecycle, diagnose a plan trajectory, nor choose which norms to observe. We show that a composer seeing only local Boolean verdicts cannot distinguish matched from mismatched transactions, then retain typed projections, the first decisive gate, and commit-time status. The distinction is summarized explicitly:

Work	Decision object	Partial observation	Cross-artifact identity	Commit-time status	Main guarantee/evaluation
Belardinelli et al. (Belardinelli et al. 2014)	artifact-centric MAS lifecycle	agent knowledge	relational artifact data	temporal lifecycle	finite abstractions and model checking
Micalizio and Torasso (Micalizio and Torasso 2014)	multi-agent plan execution	distributed action observations	dependencies between action failures	asynchronous trajectory	diagnosis of primary/secondary failures
Criado (Criado 2018)	monitored norm set	resource-bounded observations	norm/event applicability	monitor execution	monitor properties and simulations
this work	one delegated authorization transaction	local typed verifier outputs	explicit partial subject projections	atomic status/effect boundary	local-verdict theorem plus executable witnesses

Dell’Anna et al. revise conditional norms from execution traces (Dell’Anna et al. 2022), while Saisubramanian et al. mitigate side effects caused by incomplete open-world models (Saisubramanian et al. 2022). They remain relevant conceptual neighbours, but are not claimed as the closest technical precedents.

3.7 Typed verifier states and concurrent commit

The public first-party EATF (Agent Trust Framework) toolkit supplies a two-language Action Evidence Package (AEP) verification example with an author-defined decision-path oracle and native first-failure codes in TypeScript and Python (Tyche Institute 2026). The toolkit is published under the historical expansion “Agent Evidence Package”; this article uses the framework’s current canonical expansion, “Action Evidence Package”, for the

same object model, and cites the published artifact under its own recorded title. We use that result only to test representational transfer: its codes remain native and are crosswalked to a coarse shared ontology for analysis.

For revocation, the relevant systems property is linearizability: concurrent operations should appear in a single order consistent with their real-time precedence (Herlihy and Wing 1990). A status read followed by a separate effect is not an atomic operation, even when the read is fresh and signed. Our live-race laboratory therefore compares read-based profiles with a status-and-effect guard whose check and commit share one server-side linearization point. Native state appraisal uses the TPM 2.0 quote-verification interface but only over a frozen software-TPM vector (Trusted Computing Group 2026).

4 Decision-Transaction Model

The implemented record makes the cross-artifact subjects explicit through `canonical_action` and `observed_effect`. Both fields are written by the corpus builder, re-derived by the verifier through the shared `composition_rule.subjects_of` extractor, and read by the binding stage. This checks serialization and plumbing, not independent subject extraction. The record is:

```
DecisionTransaction {
  id                -- stable transaction identifier
  family            -- one of the five corpus families
  policy_vector_id  -- selects the signed policy and its evidence
  evidence_variant_id -- optional evidence override, else null
  measurement_fixture_id -- selects the measurement fixture
  measurement_fixture_source -- provenance of that fixture
  state_fixture_id  -- names the attestation-result fixture
  attestation_result { -- the embedded appraisal input
    status, issued_at, expires_at,
    observed_digest, reference_digest, profile }
  authority_case_id -- selects the typed delegation-path case
  canonical_action { -- the action the artifacts jointly authorize
    effect, resource, operation,
    granted_tools, granted_resources,
    authorised_from, authorised_until,
    measurement_profile }
  observed_effect { -- the effect as reported and appraised
    effect, resource, issued_at, measurement_profile }
  expected { -- the author-written expected label
    verdict, first_rejecting_gate, failed_layers,
    layer_results, binding_result, binding_gate }
  note -- optional prose, on the binding family only
}
```

Neither subject record is a new artifact, and neither introduces a sixth verifier. Every field is read out of an artifact that one of the five verifiers already inspects:

- `canonical_action.effect` and `canonical_action.measurement_profile` are the signed policy object's required_effect and required_measurement_profile;
- `canonical_action.resource` and `.operation` are the delegation case's terminal action;

- `canonical_action.granted_tools` and `.granted_resources` are the terminal narrowed scope, that is the grant carried by the last delegation edge, which the authority verifier has already certified to be a subset of every parent scope whenever it returns `ALLOW`;
- `canonical_action.authorised_from` and `.authorised_until` are the intersection of the per-edge validity windows, the interval in which the whole path is valid;
- `observed_effect.effect`, `.resource`, and `.issued_at` are fields of the signed evidence object whose signature the evidence layer verifies;
- `observed_effect.measurement_profile` is the profile identifier of the fixture the measurement layer actually appraised.

The corpus builder stores both records; the verifier re-derives them from the re-executed inputs and asserts field equality, reported as `subject_matches` in the shipped summary, so these are checked record fields rather than decoration. They remain corpus-supplied strings throughout, and comparing them is not attestation of anything.

Every layer returns a typed result rather than one undifferentiated Boolean, and so does the binding stage:

```

policy      in {PASS, MISSING, STALE, SUBSTITUTED, INVALID_SIGNATURE}
evidence    in {PASS, TAMPERED, REPLAY}
state       in {PASS, CONTRAINDICATED, STALE, REFERENCE_MISMATCH}
authority   in {ALLOW, DENY(gate, edge_id, edge_index)}
measurement in {PASS, FAIL_POINT, FAIL_LCB, PROFILE_MISMATCH}
binding     in {PASS, EFFECT_MISMATCH, RESOURCE_MISMATCH,
               TIME_MISMATCH, PROFILE_MISMATCH, UNDETERMINED}

```

The evaluation order is fixed, and the binding stage is last:

```
policy -> evidence -> state -> authority -> measurement -> binding
```

All five layers are appraised to preserve a typed failure set. The `first_rejecting_gate` follows the fixed order and supports deterministic localization. It is not a claim that later failures did not exist.

Two properties of the binding stage belong here because they are properties of the record; the rule itself is given in Section 5.4. First, `failed_layers` deliberately remains a list over the five artifact layers only. The binding stage never appears in it and has no row or column in the co-failure matrix, so a transaction whose `failed_layers` is empty and whose verdict is `DENY` is exactly a cross-layer denial. Second, the binding function is pure over the two subject records and is evaluated for every transaction regardless of the layer results, while the gate ladder consults its outcome only after all five layers have passed. This is not fail-open: a transaction with a failed layer is already a `DENY`, and the earlier gate is the honest localization. When a required subject field is absent — for example when a `MISSING` policy supplies no `required_effect` — the binding result is `UNDETERMINED`, which never counts as a pass.

4.1 The formal spine: local-verdict indistinguishability

Let a_P, a_E, a_S, a_A, a_M be the five artifacts and let $V_i(a_i) \in \{0, 1\}$ be their local acceptance predicates. Each accepted artifact also exposes a partial subject projection $\pi_i(a_i)$: effect, resource, interval, measurement profile, or some subset. Define

$$L(x) = \bigwedge_{i \in \{P, E, S, A, M\}} V_i(a_i)$$

and let $B(x)$ be the conjunction of the cross-artifact equality, membership, and interval constraints in Section 5.4. The strict decision is $C(x) = L(x) \wedge B(x)$. A Boolean-only composer observes $v(x) = (V_P, V_E, V_S, V_A, V_M)$ but not the projections.

The implemented B is deliberately a **partial protected-subject schema**: it binds the policy-required effect and measurement profile to the signed reported effect, the authority action/scope resource, and the report issued_at to the authority interval. It does not yet bind principal, request identifier, full action parameters/digest, state workload identity, measured model/dataset identity, actual external effect time, or commit receipt. The theorem applies when a required global relation is not observable through $v(x)$ and a matched/mismatched pair exists; it does not automatically establish every omitted field. The executable 104-case result establishes only the four implemented relations. We use “subject-bound” below only in that bounded sense. The implemented model additionally assumes a nonempty, single, linear delegation path: the terminal-grant projection reads the final edge of one path, no zero-hop fixture exists, and branching or concurrent delegation graphs are out of scope.

Theorem 1 (local-verdict indistinguishability). Suppose there are a matched transaction x^+ and a mismatched transaction x^- such that

$$v(x^+) = v(x^-) = (1, 1, 1, 1, 1), \quad B(x^+) = 1, \quad B(x^-) = 0.$$

Every deterministic composer f whose only input is $v(x)$ returns the same decision for x^+ and x^- . Consequently, it cannot both allow the matched transaction and deny the mismatched transaction.

Proof. Equality of the observed vectors gives $f(v(x^+)) = f(v(x^-))$. The required decisions differ, so either the matched transaction is falsely denied or the mismatched transaction is falsely allowed. $\square$

This is an information limitation, not a defect specific to majority voting. The corpus provides eight x^- witnesses and six matched x^+ controls, including a locally valid signed report for `ledger.write` paired with a locally valid terminal grant for `ledger.read`. Their existence is constructed evidence under the partial subject schema defined below, not a deployment-frequency estimate.

Corollary 1 (strict refinement). $C(x) \Rightarrow L(x)$, and the refinement is proper whenever a witness x^- from Theorem 1 exists. This follows directly from $C = L \wedge B$.

Proposition 2 (failure of compensating composition). Call a rule *compensating* if it is a monotone function of a subset of the predicates that can accept while at least one required predicate is false — for example, any k -of- n threshold with $k < n$, or any rule that omits a predicate entirely. For any critical predicate q omitted or outvoted by such a rule, a transaction with $q(x) = 0$ and all counted predicates equal to one is accepted by that rule and denied by C . On this corpus the four-of-five false-allow count is an algebraic identity:

$$N_{\text{FA, majority}} = N_{\text{single local failure}} + N_{\text{all local pass, binding fail}} = 23 + 8 = 31.$$

The 2 and 48 counts of the other weak ablations, and the 3/2/1 counts of the partial subject joins, depend on which fixtures the corpus contains; they are implementation observations on the frozen corpus, not algebraic properties and not rate estimates.

4.2 Decision clock and the revocation linearization point

A decision transaction has more than one relevant time. We make that explicit with a decision clock

$$\kappa(x) = (t_P, t_E, t_S, t_A, t_M, t_B, t_C),$$

where the first six components are the policy, evidence, state, authority, measurement, and binding appraisal instants and t_C is the effect-commit instant. The clock is expository notation: no shipped artifact computes $\kappa(x)$, and the live laboratories order events by the server-side linearization index ℓ alone. Each artifact contributes its own validity interval or freshness constraint. A revocation event has an effective instant t_R . Wall clocks, however, are insufficient to order concurrent requests reliably, so the live laboratory records a server-side linearization index ℓ for every status read, revocation, guarded commit, and unguarded effect.

Proposition 3 (atomic revocation safety under the laboratory model). If a revocation operation and a guarded effect commit are linearizable operations on the same credential state, and the guarded commit writes the effect only when that state is active at its linearization point, then every committed effect was active at commit. A separate status read cannot provide this guarantee: the admissible order

$$\ell_{\text{read}} < \ell_{\text{revoke}} < \ell_{\text{effect}}$$

returns an authentic active status and still commits after revocation.

Argument. The guarded operation holds the state lock across check and commit, so either it linearizes before revocation and commits while active, or revocation linearizes first and it denies. The three-operation schedule above is a counterexample for every profile whose check and effect have different linearization points. This is a property of the experimental service semantics, not a universal deployment theorem.

A dependency-free Java model checker separately exhausts the bounded models used by Theorem 1, Corollary 1, and Proposition 3. It imports no Python adapter, corpus builder, SQL oracle, expected-label module, or saved verdict stream. It enumerates both possible outputs of a Boolean-only composer on the shared all-pass vector; all 16 agreement states over effect, resource, time, and measurement profile; all six orders of one read, revoke, and commit; and all four topologically valid two-read schedules. This is implementation-diverse finite enumeration, not a proof of arbitrary programs or distributed systems. For the revocation schedules, the checker’s atomic-guard oracle encodes the guarded semantics by construction, so that branch documents the incomplete profiles’ time-of-check-to-time-of-use (TOCTOU) witnesses rather than independently establishing atomicity; the atomicity evidence is the live, process-isolated, and containerized laboratories of Sections 5.6 and 6.4.

4.3 What a Boolean verdict destroys

Let a typed verifier expose K distinct rejection states. The scalar map $s(r) = \text{REJECT}$ for every rejecting r is non-injective and erases all $K(K-1)/2$ pairwise distinctions among them. This count is independent of case frequency; entropy and best-guess accuracy additionally depend on the fixture mix. In the transfer laboratory EATF has 16 native rejection states and the composed corpus has 28, so Boolean scalarization erases $16(15)/2 + 28(27)/2 = 498$ within-corpus pairwise distinctions. The analytic crosswalk retains every native code and adds a coarse comparison class; it does not replace the code.

5 Methods

The study design is summarized before the laboratory details so that each question has one primary artifact, observation, and claim ceiling.

RQ	Primary artifacts	Oracle or invariant	Primary observation	Ceiling
RQ1: local-validity limit	16 policy/evidence vectors; SAFE metamorphics; 104 composed transactions including 14 binding cases	typed precedence, metamorphic relations, re-executed layer evaluators, matched local-verdict witnesses	policy and measurement gate sensitivity; eight all-local-pass denials; false allows of weak and partial-binding ablations	constructed data; shared-rule composition labels are analytic

RQ	Primary artifacts	Oracle or invariant	Primary observation	Ceiling
RQ2: runtime atomicity	104 TPM-bound state vectors + 64 mutations; 372 durable service cases	native TPM verification and frozen state oracle; transactional linearization and idempotency	104/104 state classes, 64/64 mutation rejects, 104/104 recompositions; safety, recovery, and duplicate-effect invariants	eight software-TPM roots and separate services, but one x86_64 physical host
RQ3: typed transfer	21 EATF rows + 104 transaction rows	native codes preserved by total crosswalk; separate SQL execution and mapping mutation	code coverage, crosswalk sensitivity, and scalarization loss	author-defined analytic mapping; no external semantic validation or interoperability

5.1 Policy-Version x Evidence Replay

The name **Policy-Version x Evidence Replay** reserves “state” for the runtime predicate S and distinguishes it from policy and evidence conditions such as missing, stale, substituted, tampered, and replayed.

We generated a deterministic 16-vector corpus: the full factorial

$$\{\text{correct, missing, stale, substituted}\} \times \{\text{good, tampered, replayed}\},$$

plus four gate-isolation vectors, each carrying good evidence and failing exactly one strict policy gate:

- **version-only**: correct policy, version 1, active interval;
- **freshness-only**: correct policy, version 2, expired interval;
- **digest-only**: correct identity, version, and interval, one semantic field changed and re-signed, so only the digest comparison rejects;
- **invalid policy signature**: the policy object tampered after signing.

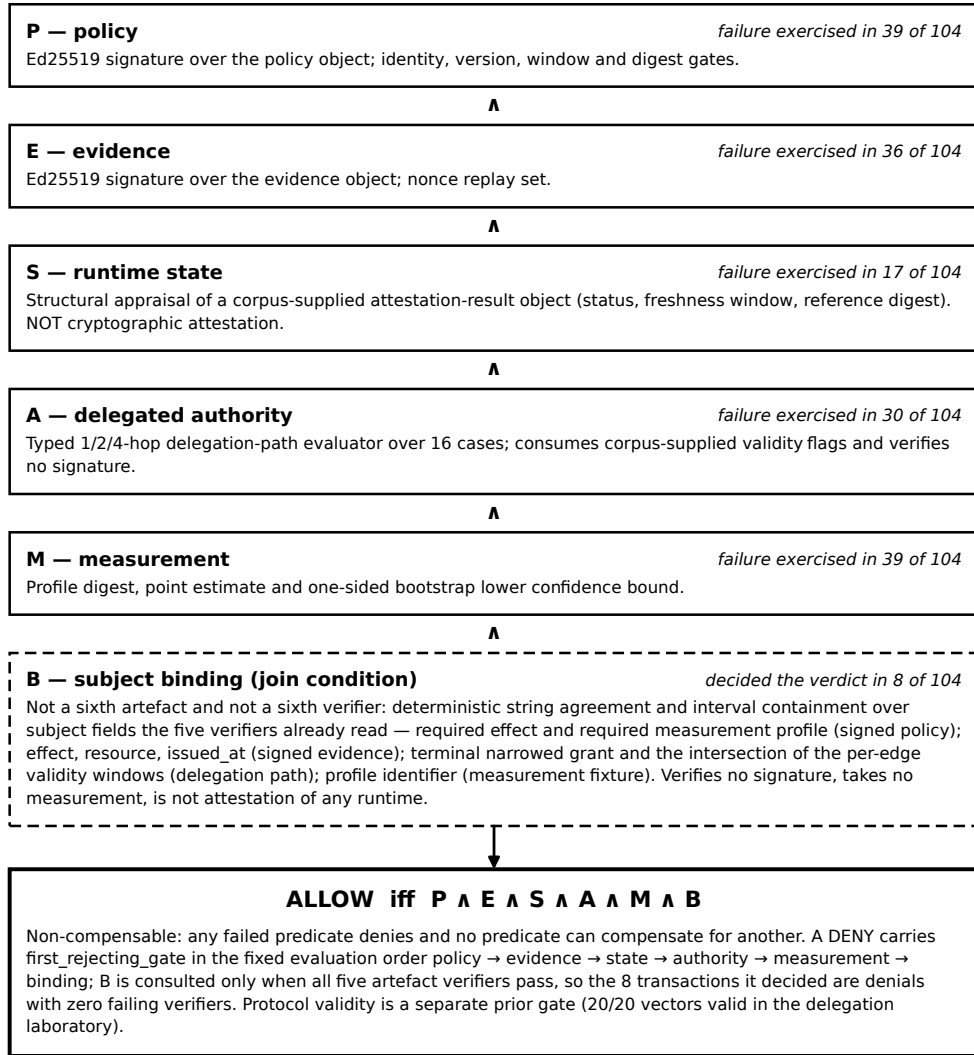
The four isolation vectors separate the version and freshness gates, make the digest branch reachable, and exercise policy `INVALID_SIGNATURE`. Each strict gate (signature, identity, version, time, digest) is therefore exercised in isolation by at least one vector.

Policies and evidence are signed with a deterministic Ed25519 test key. The key is public corpus material. The factorial vectors instantiate:

- **correct policy**: expected identity, version 2, active interval, and exact unsigned-object digest;
- **missing policy**: no policy object;
- **stale policy**: valid signature, version 1, and an expired interval;
- **substituted policy**: valid signature but a different policy identity and required effect;
- **good evidence**: valid signed effect with an unseen nonce;
- **tampered evidence**: effect changed after signing;
- **replayed evidence**: valid signature with a pre-seen nonce.

We compare the strict verifier with two experimental ablations:

1. signed-policy presence only;
2. missing-policy fail-open plus signed-policy presence.



Solid boxes are the five artefact verifiers; the dashed box is the join condition over their subjects. Counts describe the designed 104-transaction corpus with author-written expected labels; they are not rates and do not estimate prevalence in any deployed system. State appraisal is structural, not cryptographic attestation; authority flags are corpus-supplied, not verified signatures; the binding stage compares corpus-supplied subject strings and is not attestation of any runtime.

Fig. 1. The composition rule. Each of the five predicate nodes is annotated with the mechanism that establishes it in this artifact packet and with how often that layer fails across the 104 frozen transactions; the binding stage is not a layer and is annotated with the number of transactions whose verdict it decided. Counts describe corpus composition, not any deployed system; the state mechanism is structural appraisal, and the binding stage compares corpus-supplied subject strings (Section 2 ceiling).

A second, relational re-composition code path recomposes the verified policy and evidence layer states in SQL. It consumes the per-layer typed results emitted by the Python evaluator and re-derives only the conjunction and the first-rejecting-gate label; it does not reimplement Ed25519 and is not independent validation.

The experiment is motivated by a recorded Veraison freshness run in which Veraison correctly rejected a replay while the intended policy was active; the missing-policy and stale-policy fail-open cases occurred in this programme’s own provisioning harness, not in Veraison. That record is an internal companion artifact; a public deposit is pending.

5.2 SAFE Metric Metamorphics

This laboratory consumes a frozen corpus read-only.

For positive vectors a , b , and c , we evaluated tensor means of:

$$M_A(x, y, z) = \frac{x + y + z}{3},$$

$$M_G(x, y, z) = (xyz)^{1/3},$$

$$M_R(x, y, z) = \sqrt{\frac{x^2 + y^2 + z^2}{3}}.$$

The arithmetic and geometric tensor means have the factorized identities:

$$V_A = \frac{\bar{a} + \bar{b} + \bar{c}}{3},$$

$$V_G = \overline{a^{1/3} b^{1/3} c^{1/3}}.$$

The suite uses a pure-Python loop implementation and a NumPy tensor transliteration of the same specification. Agreement between them excludes vectorisation and floating-point error, not specification error. The method is metamorphic testing (T. Y. Chen et al. 1998; Segura et al. 2016): each check states a relation that must hold between outputs rather than an oracle for one output. The fifteen checks divide into nine implementation identities (two-implementation agreement for the three integration operators and for TOPSIS, the arithmetic and geometric factorizations, the ordering $V_G \leq V_A \leq V_R$, and axis and within-axis permutation invariance) and six informative checks (tensor-versus-aligned-diagonal non-equivalence, profile-digest sensitivity under six semantic mutations, dynamic-reference TOPSIS rank behaviour, fixed-reference TOPSIS rank behaviour, measurement-fixture typing over four fixtures, and the sampling-overlap boundary fixture).

The synthetic confidence fixtures use shared three-column samples. For each fixture we compute a point estimate and a one-sided bootstrap lower bound $LCB_{1-\alpha}(C)$. The strict measurement rule requires:

$$\text{profile digest matches} \wedge \widehat{C} \geq \tau \wedge LCB_{1-\alpha}(C) \geq \tau.$$

The bootstrap is fully disclosed here because the boundary decision depends on it: the lower bound is the plain percentile method, 5,000 deterministic replicates, NumPy PCG64 generators seeded per dataset at 101, 102, and 103, with the resampling unit being the row of the shared sample. The fixture sample sizes are $n = 30$ (supported), $n = 18$ (point-only boundary), $n = 30$ (unsupported), and $n = 30$ (altered profile, which reuses the supported dataset). No bias-corrected or accelerated (BCa) interval was computed – a limitation, and the percentile method’s small-sample bias at $n = 18$ is exactly where the boundary fixture sits. A companion sensitivity sweep over five seeds and 2,000/5,000/10,000 replicates, using the same percentile bootstrap, leaves every fixture’s confidence outcome unchanged at all 15 grid points (per-fixture lower-bound spread at most 0.0017) and preserves the

overlap fixture’s joint-versus-independent decision flip at 15/15 grid points; the archived sweep is part of the artifact. The evaluator applies its gates in a fixed first-fail order symmetric with the binding ladder of Section 5.4: profile digest, then point threshold, then lower confidence bound. Bootstrap validity is also dependence-sensitive (Cameron et al. 2008; Künsch 1989; Politis and Romano 1994), which motivates the overlap fixture below. The fixtures were designed to test boundary behavior and are not measurements of a deployed model.

The TOPSIS fixture contains alternatives A, B, and C plus one added alternative. Dynamic observed-reference TOPSIS recomputes normalization, ideal, and anti-ideal points from the current set. The fixed-reference variant uses a declared normalized 0–1 reference.

5.3 Authority-to-effect evaluator

The authority layer reuses the typed delegation-path suite from the protocol-valid-but-unauthorized laboratory of this packet. It constructs 1/2/4-hop paths and evaluates:

- lineage and role transitions;
- issuer-signature flags;
- permitted subdelegation;
- monotone scope;
- status and freshness;
- terminal action coverage;
- effect time and action binding.

The suite contains 16 typed cases. Four isolate `protocol.valid`, `edge.signature`, `effect.native_validity`, and `effect.time_binding`, making each branch reachable in both the Python evaluator and the relational oracle.

Three of them — `D1_protocol_invalid`, `D2_signature_invalid_second_edge`, and `D1_receipt_native_invalid` — flip the corpus-supplied Boolean the gate under test reads. That is *branch coverage* in two implementations at once, not verification of the property the branch names: no signature is checked and no protocol run is validated. What it rules out is a branch that is unreachable, and therefore silently untested, in both implementations. Only `D1_receipt_effect_time_outside_window` exercises a computed comparison, moving the receipt’s effect time outside the delegation window.

The suite reports 16/16 expected typed tuples, 13/13 first-invalid-stage matches on the negative cases, and 8/8 delegation-edge ID and index matches under an edge-gate definition that counts only cases rejected by an edge.* gate. The same laboratory also retains a 20-vector semantic corpus (20/20) and a seven-case hybrid adapter over recorded A2A evidence (7/7).

The cryptographic component of this layer is stipulated, not computed: the fields `issuer_signature_valid`, `native_evidence_valid`, and `protocol_valid` are corpus-supplied experimental flags, and no cryptography executes anywhere in the authority laboratory. The result schema uses explicit coverage names such as `native_object_flags_true`, and the disclaimer is embedded in the result JSON, not only here.

The delegation-path suite agrees 16/16 with an independent relational SQL implementation. We keep the word “independent” for this oracle alone, and it differs in kind from the composition-level SQL of Sections 5.1 and 5.4: it is a 233-line relational re-derivation that starts from the raw case objects, normalises edges, scopes, and receipts into tables, computes a violation set, and ranks violations with a window function, a set-then-rank formulation confronting the Python evaluator’s imperative short-circuit. The case expansion and expected labels remain shared with the Python entry point, so even this agreement is implementation diversity over verdict derivation, not independent validation of the labels.

5.4 Composed frozen corpus

The composed corpus contains 104 transactions:

Family	Transactions	Purpose
policy/evidence/measurement factorial	36	cross the 12 signed replay vectors with three measurement outcomes
state-authority-measurement cube	8	enumerate all binary pass/fail combinations of those three layers
authority path reuse	16	re-execute all 16 typed 1/2/4-hop cases with other layers valid
cross-layer joins	30	10 layer pairs x 2 two-fault joins, 4 triples, 1 quadruple, 1 all-five failure, and 4 matched all-valid lookalike positives
cross-layer binding	14	8 transactions in which all five layers pass and the subject records disagree, each paired with a matched all-valid lookalike (6 distinct controls)

The cross-layer-joins and cross-layer-binding families do two different jobs that must not be conflated under the word “join”.

The cross-layer-joins family populates the co-failure structure. It fills every off-diagonal cell of the 5 x 5 co-failure matrix, with a corpus-wide minimum off-diagonal count of 5, and its four matched all-valid lookalike positives ensure that universal refusal cannot masquerade as correctness. A populated co-failure cell is a co-occurrence of two independently caused layer failures in one transaction. It is not a join in the relational sense, and it does not test whether the layers are describing the same action.

The cross-layer-binding family tests agreement over the implemented effect, resource, report-time, and measurement-profile coordinates; it does not test the omitted state/workload, principal, model, or commit-receipt coordinates. Each of its 8 denials is a transaction in which every one of the five component verifiers returns a pass and the composition denies because one implemented coordinate disagrees; each is paired with a matched all-valid lookalike that is allowed, so the denials are not universal refusal. Its evidence objects are owned by this laboratory and signed with the policy laboratory’s own test key and signing function, imported rather than transcribed; every one carries a fresh nonce and a valid signature, so the unmodified evidence evaluator returns PASS on all of them and they differ only in the subject fields the binding stage reads. Its two extra measurement fixtures are likewise owned by this laboratory, in the SAFE fixture schema, appraised by the unmodified SAFE evaluator, which passes both; they differ from each other only in profile identity, and they exist because the signed policy cannot be varied without changing its digest, which would make the policy layer report SUBSTITUTED. Each transaction’s measurement_fixture_source records which population its fixture came from. All these constraints are enforced as deterministic assertions at corpus build time and re-checked by the verifier’s exit gate.

The binding stage itself is a single pure function over the two subject records, applied in a fixed rule order, first failing rule winning:

0. any required subject field absent → UNDETERMINED
1. signed reported effect differs from the policy-required effect,
or is not in the terminal granted tools → EFFECT_MISMATCH
2. reported resource differs from the canonical action resource,
or is not in the terminal granted resources → RESOURCE_MISMATCH
3. issued_at outside the authorised window → TIME_MISMATCH
4. measurement profile differs from the
policy-required profile → PROFILE_MISMATCH
5. otherwise → PASS

Each non-PASS outcome maps to the correspondingly named gate in the binding.* namespace. Timestamps are compared lexicographically, which is sound only because every timestamp in this artifact is fixed-width UTC Z form; both window boundaries are inclusive, matching the state adapter and the policy validity window. The time test is containment and never equality. The three decision-relevant time constants in the artifact — the policy and state decision time 2026-07-25T21:30:00Z, the authority effect time 2026-07-25T20:00:00Z, and the evidence issue time 2026-07-25T21:29:30Z — are unrelated. The main corpus therefore cannot instantiate the decision clock of Section 4.2 from those constants. The live companion does instantiate its critical order with server-side linearization indices.

The stage is placed in the composition rather than inside a layer because a predicate over four artifacts has no local home. Putting it inside the evidence evaluator would make the evidence column of the co-failure matrix partly a copy of the policy column, and could not express the resource, time, or profile mismatches at all; putting it inside the authority evaluator would be cross-layer analysis wearing a single-layer costume and would make the composition rule vacuous. Only the join can see all four subjects, so only the join can reject on their disagreement.

A standalone acceptance test rebuilds the Section 1 counterexample and its lookalike from first principles and does not read the frozen corpus, so the result cannot be satisfied by a corpus entry carrying a convenient label. It asserts, and prints, that the shipped evidence evaluator still returns PASS on the WRITE receipt — the root cause is unchanged, and the fix is at the join and not in the layer — that all five artifact verifiers pass, that the binding result is EFFECT_MISMATCH, that the verdict is DENY at gate binding.effect, and that the lookalike is ALLOW at gate verified.

Two provenance properties are central and are declared inside the corpus and every generated summary. First, the conjunction, binding stage, and first-rejecting-gate ladder are centralized in one shared composition-rule module imported by both the corpus builder and verifier. Consequently, composition-level expected labels are generated by the same composition rule the verifier imports, so agreement at that level is analytic. Per-layer agreement is the non-analytic execution check; subject-record agreement is a same-function round trip. Second, transactions embed attestation_result objects, and the verifier calls a state adapter at the fixed decision time 2026-07-25T21:30:00Z (asserted equal to the policy corpus decision time) rather than comparing a corpus string. The adapter is a structural appraisal of the supplied object: a status gate, an inclusive freshness window, and a reference-digest comparison, in a fixed order, emitting PASS, CONTRAINDICATED, STALE, or REFERENCE_MISMATCH, and failing closed on missing fields. It performs no cryptographic verification, involves no TPM and no Veraison, and is not remote attestation; the disclaimer travels inside the result records. A 24-case adapter self-test runs as the first gate of the laboratory’s entry point.

Corpus construction records the SHA-256 of each source corpus. The composed verifier verifies those hashes and re-executes the three source evaluators per transaction through importlib; it does not copy stored verdicts. The SAFE source corpus is consumed read-only from its frozen source.

5.5 Ablations

We compare the strict composition with:

- **artifact validity / policy presence:** signed-policy presence and evidence integrity, ignoring policy identity, state, authority, measurement, and binding;
- **point-only measurement:** strict policy, evidence, state, authority, and binding, but $\widehat{C} \geq \tau$ without confidence or profile enforcement;
- **four-of-five majority:** allow when any four of the five artifact layers pass;
- **effect/resource binding:** require all five local verifiers plus effect and resource agreement, but omit time and measurement-profile agreement;

- **effect/resource + time:** add authorized-window containment, but omit measurement-profile agreement; and
- **effect/resource + profile:** add measurement-profile agreement, but omit authorized-window containment.

The two compensating ablations differ in what they remove. The point-only profile replaces exactly one thing, the measurement criterion, and retains the binding stage; that is what makes it an ablation of one thing, and it is why the binding function is evaluated for every transaction rather than only when the layers pass. The four-of-five majority is by construction also an ablation *of the binding stage*: it votes over the five artifact verifiers, and a stage that is not one of them cannot appear in the vote.

These are intentionally incomplete experimental profiles, not representations of named products. The latter three are credible subject-aware schema ablations rather than local-validity strawmen: each checks a useful subset of the join and omits a different coordinate. Their false-allow counts are reported in Section 6.3, together with the analytic identity that governs the majority profile, because the aggregate counts are functions of corpus design.

5.6 Companion evidence-class upgrades

The main 104-transaction corpus treats its supplied experimental authority-validity flags as controlled factorial inputs. Nine companion laboratories change evidence class without silently reinterpreting that corpus; a tenth, the standalone finite model check, accompanies the formal apparatus (Section 4.2).

Native signed authority/effect adapter. Eight deterministic fixtures use real Ed25519 verification for every delegation edge and the effect receipt. The verifier checks parent digest, principal and role lineage, permission, status, freshness, scope attenuation, terminal action scope, receipt issuer, chain digest, effect time, and action binding. One fixture passes; seven isolate a bad edge signature, bad receipt signature, signed action mismatch, expired edge, signed wrong lineage, signed scope amplification, and signed effect-time mismatch. The builder uses deterministic test-key seeds and persists public keys only. This is a Tyche experimental profile, not conformance with WAVE, JEDI, OAuth, DID, or another external credential format.

Mutation-generated transactions. Starting from one independently passing transaction, a deterministic builder applies 37 single faults, one at a time (one baseline plus 37 generated cases, the 38 frozen pairs), across policy rules P1–P5, evidence rules E1–E2, state rules S1–S3, authority rules A0–A19, and measurement rules M1–M3. The transaction packet, both evaluator sources, and both unsealed outputs were hashed before comparison with the author-written mutation oracle. A second packet composes 12 cases with two to five simultaneous faults, including two within-layer precedence cases. The freeze establishes ordering, not label truth: generator, oracle, and labels remain programme-internal.

Prospective signed-revocation races. Thirteen scheduled event sequences place signed status snapshots before appraisal, between appraisal and commit, at the commit instant, and after commit, adding stale, replayed, substituted, unavailable, re-issued, out-of-window, and bad-signature status cases. The strict rule verifies signed credential and status objects, uses the half-open credential window `valid_from <= t < valid_until`, requires fresh status at appraisal and commit with a non-decreasing signed sequence number, and treats revocation at the commit instant as effective. Independent Python and Node implementations evaluate the frozen packet; three ablations omit commit rechecking, fail open on missing commit status, or ignore sequence monotonicity.

Native runtime-attestation appraisal. The lab embeds the public frozen Tyche `aep-pcr16-vector` at commit `2d22013987e1e98f8f9132832c9db2b967035945`, including its RSA attestation public key, `TPMS_ATTEST`, TPM signature, and PCR-values blob. The verifier recomputes

```
PCR16 = SHA256(0x00*32 || outcome_digest_bytes)
preimage = "tyche.aep.qual.v1|capsule=" || capsule_hex
```

```

753         || "|outcome=sha256:" || outcome_hex
754         || "|nonce=" || challenge_hex
755     qual_hex = HEX(SHA256(ASCII(preimage)))
756     checkquote_q = HEX(ASCII(qual_hex))

```

and invokes `tpm2_checkquote` with the last value, matching the frozen vector's ASCII-hex convention. One affirming case and six author-written cases, hash-pinned before the recorded invocation, cover outcome substitution, fresh-challenge replay, capsule substitution, declared-PCR substitution, quote-message tampering, and quote-signature tampering. The source vector was produced by `swtpm`; no hardware endorsement or live workload is inferred.

Live concurrent revocation service. A `ThreadingHTTPServer` on an ephemeral loopback port linearizes signed status reads, revocations, guarded commits, and unguarded effects under one lock, assigning each operation an increasing index; every status-bearing response is Ed25519-signed and client-verified. Four profiles are crossed with five revocation placements plus 64 same-barrier revocation/commit trials each. `atomic_guard` checks status and commits the effect within one locked operation; `double_read`, `single_read`, and `t1l_cache` make an unguarded effect after a fresh, appraisal-time, or cached status (the 50 ms cache with a forced 75 ms expiry deterministically separates `t1l_cache` from `single_read`). A false allow is defined by the service record itself: the client accepted and the credential status at the effect's linearization point was revoked.

Native state transaction overlay. The retained structural objects define the frozen source oracle, but no status field enters the native adapter. For each transaction, a subject-derived capsule identifier binds the source-corpus hash, transaction identifier, canonical action, and observed effect. PCR 16 binds a transaction-specific runtime measurement, while qualifying data binds the capsule, measurement, signed validity interval, and unique 32-byte challenge. Eight freshly initialized `swtpm` roots issue 104 quotes: four RSA and four ECC attestation keys (AKs), thirteen transactions per root. After `tpm2_checkquote`, PCR, challenge, and subject checks, a frozen appraisal policy applies contraindication, signed-window, and reference-measurement rules in the same precedence as the source state vocabulary. The verifier then re-runs the shared composition rule with only the state result replaced. One representative vector per root receives eight predeclared mutations: quote message, signature, PCR blob, challenge, transaction identifier, source corpus hash, measurement, and signed window.

An independent compiled verifier reprocesses the same frozen native packet in OpenJDK 21. It imports and executes none of the Python adapter, structural corpus, corpus builder, expected-label module, or saved mutation generator. It implements its own strict JSON parser, canonical transcript, SHA-256 and PCR derivations, appraisal precedence, and mutation construction, then invokes `tpm2_checkquote` directly. `jdeps` reports only `java.base`. The comparison therefore tests implementation diversity against the frozen packet, while remaining author-operated, same-host, and dependent on the same packet and TPM2 command-line verifier.

Process-isolated durable revocation service. Status and effect run as separate OS processes with separate TCP listeners. They coordinate through a SQLite append-only event log; every status read, revocation, and effect receives a durable linearization index and a signed response. The guarded effect checks status and inserts an idempotency-keyed effect in one `BEGIN IMMEDIATE` transaction. Simultaneous revoker and committer contenders run in separate client processes. In addition to the five scheduled placements and 64 same-release trials per profile, the runner disconnects after commit request but before response consumption, retries, and in a second fault class kills and restarts the effect service before retry. The primary safety oracle is the status stored in the effect transaction; the effect table supplies the duplicate-effect oracle. The experiment remains same-host and uses SQLite as a centralized linearization mechanism.

Containerized durable-revocation replay. A pinned offline image runs a status service, two interchangeable effect-service instances, a deterministic response-fault proxy, and the runner as five read-only containers over an internal bridge, with a named-volume SQLite write-ahead-log store as the centralized linearization mechanism.

The proxy injects request loss before forwarding and response loss after durable commit; recovery retries through the second effect instance and replays both allowed and denied idempotency keys. A separate saved-result verifier rechecks every signature, reconstructs the false-allow predicate from stored effect-time state, and checks unique decision/effect keys, contiguous event identifiers, database integrity, and distinct signed service hostnames. Container separation changes process, filesystem, network-namespace, and service-instance boundaries — not the physical host, kernel, datastore trust domain, or operator.

Cross-ecosystem typed transfer. A pinned, field-minimized snapshot of the public EATF decision-path result supplies 21 rows (two accepts, 19 rejects covering 16 native rejection codes, exact TypeScript/Python agreement). Every native code and first gate from the 104 transactions must appear in an explicit total crosswalk to eight coarse analytic classes; the builder exits non-zero on an unmapped code or a source-oracle mismatch. Native codes remain the primary result. A separate SQLite implementation consumes the frozen JSON mapping and performs a leave-one-out plus wrong-class mutation for every entry, testing execution-path agreement, totality, and sensitivity — not the semantic correctness of the ontology.

A separate blind semantic-validation kit converts the 46 native states into opaque mapping tasks with nine class definitions, a NO_FIT option, and a response schema, excluding the author mapping (hash-sealed outside the archive) and all result fields. A coordinator validator, preregistered unweighted-kappa analysis with a 0.60 stop rule, and fail-closed checks for duplicates, low agreement, residual NO_FIT, and unresolved taxonomy-defect are frozen before any dispatch; its self-tests pass with `author_mapping_opened=false`. The kit remains undispatched and therefore supplies no independent semantic evidence.

The human-labelling experiment is deliberately separate. Its neutral packet is a label-isomorphic 104-case transformation of the composed corpus: the builder replaces mnemonic identifiers with opaque aliases, re-signs policy and evidence objects under a packet-specific deterministic Ed25519 key, preserves invalid signatures and replay membership, and re-evaluates every typed tuple (104/104 preserved; the known-token audit reports no remaining mnemonic match). The response schema, sealed author labels, adjudication protocol, and a pre-registered analysis program (verdict kappa with a fixed 10,000-replicate paired bootstrap, an uncollapsed gate confusion matrix, and the 0.60 stop gate enforced without opening author labels) are prepared; the packet is undispatched, and no external label, adjudicated result, or agreement statistic exists.

6 Results

6.1 Policy state changed the appraisal surface

The strict verifier matched 16/16 expected tuples and allowed only the correct-policy/good-evidence cell. The signed-policy presence profile matched 11/16 with five false allows: good evidence under stale, substituted, version-only, freshness-only, and digest-only policies. The missing-policy fail-open profile matched 10/16 and additionally falsely allowed good evidence with no policy, six false allows in total. Both baselines still validate signatures, so the tampered-policy vector is denied by all three profiles. The SQL re-composition matched the Python verdict and rejecting gate 16/16, including execution of the `policy.signature` branch.

Profile	Expected matches	False allows
strict identity/version/digest	16/16	0
signed-policy presence only	11/16	5
missing-policy fail-open	10/16	6

All counts are counts on this designed corpus, and the 16/16 needs the same analytic/substantive split that Section 6.3 applies to the composed layer. The typed per-layer results are hand-written from each declared vector condition, and recovering that condition from the signed bytes is what the verifier must do, so per-layer agreement

is the substantive check. The expected verdict and `first_rejecting_gate`, by contrast, are machine-generated by a two-layer precedence ladder that `run.py` re-implements when it composes; agreement on those two fields is agreement between two transcriptions of one small function — weaker than the composed laboratory, where a single shared module makes the corresponding agreement outright analytic, and stronger than nothing, because two transcriptions can in principle disagree. It is not evidence that the precedence order is correct.

6.2 Metamorphic and confidence results

All 15 SAFE checks passed: the nine implementation identities passed as required (they cannot fail absent a coding defect), and the six informative checks also passed. The integrated frozen values were:

Operator	Tensor volume
geometric	0.7665940656
arithmetic	0.7691666667
RMS	0.7716991533

The harness accepts two-implementation agreement at a threshold of 10^{-12} ; the observed gaps were smaller: exactly 0.0 for the three integration values and at most 1.1×10^{-16} (one unit in the last place) across the TOPSIS closeness scores. Six of six profile mutations changed the profile digest; these six mutations do not establish that the profile field set is complete.

The confidence fixtures produced:

Fixture	Point	5% bootstrap LCB	Threshold	Result
supported (n = 30)	0.825377	0.811554	0.75	PASS
point-only boundary (n = 18)	0.764537	0.732088	0.75	FAIL_LCB
unsupported (n = 30)	0.682263	0.665874	0.75	FAIL_POINT
altered profile (n = 30)	0.825377	0.811554	0.75	PROFILE_MISMATCH

A point-only gate falsely allowed two fixtures, and the two false allows are different phenomena: one confidence-gate false allow (the boundary fixture, whose point estimate clears the threshold while its lower bound does not) and one profile-gate false allow (the altered-profile fixture, which the point-only gate admits because it never checks the profile digest).

In the overlap boundary fixture the threshold 0.737 was chosen by construction to sit between the two computed bounds: joint resampling of the shared columns produced LCB 0.732088 while incorrectly resampling them as independent produced 0.740781, so the decision changes at that threshold. This is a designed counterexample demonstrating existence, not a finding about any data-generating process; the dependence-sensitivity of bootstrap intervals is established statistics (Cameron et al. 2008; Künsch 1989; Politis and Romano 1994).

Dynamic-reference TOPSIS changed the relative order of the original three alternatives from $A > C > B$ to $A > B > C$ after the fourth alternative was added. Under the fixed 0–1 reference the base ordering was already $A > B > C$ and was unchanged by the addition; the two reference schemes therefore disagree on the base ranking as well as on its stability, and the fixed scheme is not a control for the reversal but a different ranking function that is stable in this fixture. Rank reversal under reference-set change is a known property of TOPSIS and of normalisation-based multi-criteria methods generally (Aires and Ferreira 2018; García-Cascales and Lamata 2012; Wang and Luo 2009); the fixture reproduces the phenomenon inside this evidence profile, it does not discover it.

6.3 Integrated transaction results

The composed verifier reproduced all 104 expected typed tuples. That sentence requires its provenance statement before any interpretation: composition-level expected labels are generated by the corpus builder using the same shared composition-rule module the verifier imports, so agreement on the conjunction, the binding outcome, and the gate label is analytic, a property of one function applied twice. The non-analytic execution content of the 104/104 is the per-layer agreement: the re-executed policy, evidence, and measurement evaluators, the delegation-path evaluator, and the state adapter each reproducing the per-layer expectation carried by their source corpora. The subject-record agreement, also 104/104, is the stored canonical_action and observed_effect records agreeing field by field with the records re-derived from the artifacts at verification time, but both paths call `composition_rule.subjects_of`. It therefore checks serialization/plumbing and ensures the binding stage reads source fields rather than a stored verdict; it does not independently validate subject extraction.

Metric	Result
exact expected tuples (composition level, analytic)	104/104
per-layer expected agreement (substantive)	104/104
subject-record agreement, same-function round trip (plumbing)	104/104
allows	15
denies	89
cross-layer denials: all five verifiers pass, composition denies	8
relational re-composition agreement	104/104
relational binding-outcome agreement (re-derived, not echoed)	104/104
source-corpus hash closure	PASS

The cross-layer-denial row is the article's strongest single number, and it is structural: a composition rule that consumes only the five scalar layer results has no cross-layer argument, so a transaction of this class cannot even be represented, let alone denied. The corpus populates the class. The 8 transactions are:

`XLB_effect_write_two_hop`
`XLB_effect_write_other_resource_two_hop`
`XLB_effect_export_four_hop`
`XLB_resource_other_one_hop`
`XLB_resource_other_two_hop`
`XLB_issued_before_window_two_hop`
`XLB_issued_after_window_two_hop`
`XLB_measurement_profile_two_hop`

Each carries `failed_layers == []` and verdict DENY. The second, `XLB_effect_write_other_resource_two_hop`, is the counterexample this article opens with, verbatim: a WRITE effect on invoice-999 under a path that permits only READ on invoice-123. Its sibling `XLB_effect_write_two_hop` isolates the effect alone, on the authorized resource, and `XLB_effect_export_four_hop` reproduces the same seam at four hops, so the result is not a property of path length.

The relational re-composition rows are a second code path, not independent validation: the SQL oracle consumes the per-layer typed results and the subject strings emitted by the Python evaluator and re-derives the conjunction, the binding outcome, and the first-rejecting-gate label. Because it re-derives the binding rules rather than echoing them, a divergence between the two implementations of those rules would be detectable; the subject strings it compares still come from the Python evaluator, so this remains a transcription check between

8 of 104 transactions: every artefact verifier passes and the composition denies.

All 14 transactions of the `cross_layer_binding` family. In every row below the five artefact verifiers return policy PASS, evidence PASS, state PASS, authority ALLOW and measurement PASS, and the failed-layer set is empty; only agreement over the implemented effect/resource/report-time/profile coordinates varies. The binding stage is the join condition of the composition rule, not a sixth artefact verifier.

denied at the binding gate — 8 of 14		binding result	verdict
effect_write_two_hop	reported ledger.write on invoice-123 · policy requires ledger.read · terminal grant [ledger.read] on [invoice-123]	EFFECT_MISMATCH	DENY
effect_write_other_resource_two_hop	reported ledger.write on invoice-999 · policy requires ledger.read · terminal grant [ledger.read] on [invoice-123]	EFFECT_MISMATCH	DENY
effect_export_four_hop	reported ledger.export on invoice-123 · policy requires ledger.read · terminal grant [ledger.read] on [invoice-123]	EFFECT_MISMATCH	DENY
resource_other_two_hop	reported resource invoice-999 · authorised resource invoice-123 · terminal grant [invoice-123]	RESOURCE_MISMATCH	DENY
resource_other_one_hop	reported resource invoice-999 · authorised resource invoice-123 · terminal grant [invoice-123]	RESOURCE_MISMATCH	DENY
issued_after_window_two_hop	reported issued_at 2026-07-26T00:00:01Z · path valid [2026-07-25T00:00:00Z, 2026-07-26T00:00:00Z]	TIME_MISMATCH	DENY
issued_before_window_two_hop	reported issued_at 2026-07-24T23:59:59Z · path valid [2026-07-25T00:00:00Z, 2026-07-26T00:00:00Z]	TIME_MISMATCH	DENY
measurement_profile_two_hop	appraised under safe-profile-altered-v1 · policy requires safe-profile-main-v1	PROFILE_MISMATCH	DENY
matched controls — same construction, agreeing implemented coordinates		6 of 6 ALLOW	
CONTROL_read_two_hop · CONTROL_read_one_hop · CONTROL_read_four_hop · CONTROL_window_lower_two_hop CONTROL_window_upper_two_hop · CONTROL_main_profile_two_hop			

The binding stage is deterministic string agreement and interval containment over subject fields read out of corpus-supplied artefacts: it verifies no signature, takes no measurement, contacts no attestation service and is not attestation of any runtime. Its denials are denials on a designed corpus with author-written expected labels — coverage, not a rate, and not an estimate of how often such disagreement arises in any deployed system. Both window boundaries are inclusive, which is what the two boundary controls exercise. The controls are matched by construction and differ from the denials only in the reported subject fields, so a verifier that refused everything would fail them.

Fig. 2. The cross-layer denial class: the 8 transactions in which every artifact verifier passes — policy PASS, evidence PASS, state PASS, authority ALLOW, measurement PASS, empty failed-layer set — and the composition still denies, each with the disagreeing subject pair and the typed binding result that fired; beneath them, the 6 matched lookalike controls of the same family, all allowed, without which the denials would be consistent with a verifier that refuses everything. Coverage counts on a designed corpus (Section 2 ceiling).

two same-author implementations. It verifies no signature, no bootstrap replicate, no delegation edge, and no attestation object.

The ablations produced, in aggregate and per family:

Profile	Expected matches	False allows
strict typed composition	104/104	0
point-only measurement	102/104	2
four-of-five majority	73/104	31
artifact validity / policy presence	56/104	48
effect/resource partial binding	101/104	3
effect/resource + profile	102/104	2
effect/resource + time	103/104	1

Here A and D are strict allows and denies; FA is a false allow relative to strict composition.

Family	n	A	D	point FA	4/5 FA	validity FA
policy/evidence/measurement factorial	36	1	35	1	7	8
state-authority-measurement cube	8	1	7	1	3	7
authority path reuse	16	3	13	0	13	13
cross-layer joins	30	4	26	0	0	12
cross-layer binding	14	6	8	0	8	8
total	104	15	89	2	31	48

All three partial-binding errors occur in the 14-case cross-layer-binding family, where their omitted coordinates can be isolated:

Partial subject join	False-allowed mismatch class	False allows
effect/resource only	two time + one profile	3
effect/resource + profile	two time	2
effect/resource + time	one profile	1
full implemented effect/resource/time/profile binding	none	0

This ladder is the more demanding comparison: every row already requires all five local verifiers and some subject agreement. It shows the marginal coverage supplied by the implemented time and profile coordinates on the designed witnesses; it does not establish that these are the only coordinates a deployed transaction must bind.

On the 90-transaction base subset, the artifact-validity and four-of-five ablations yield 40 and 23 false allows. The point-only ablation yields 2 because the binding stage catches `F_correct_good_profile_mismatch` as `binding.measurement_profile`. The measurement evaluator's point-only result ignores the profile gate, while the binding stage rejects the profile-substituted appraisal. This follows from the ablation definition in Section 5.5: the point-only profile replaces the measurement criterion and keeps the binding stage.

A four-of-five majority allows a transaction if and only if at most one of the five artifact verifiers fails. Once denials with *zero* failing verifiers exist, its false allows are the single-fault denies **union** the cross-layer denials,

31 = 23 + 8, verified as exact set equality on transaction identifiers. The single-fault layer distribution is authority 14, policy 3, measurement 3, evidence 2, state 1. All 8 cross-layer denials are majority false allows because a vote over the five artifact verifiers cannot represent them at all — there is nothing for it to vote against.

The artifact-validity profile inspects only policy presence and evidence integrity, so it necessarily false-allows every transaction whose faults lie entirely in state, authority, measurement, or the binding: that is 13 of 13 authority-reuse denials, 7 of 7 cube denials, and 8 of 8 cross-layer binding denials. The correct reading of the whole comparison is a proof by construction, that a compensating or incomplete rule cannot preserve a non-compensable specification, together with a corpus-coverage statistic; it is not a measurement of external systems.

Layer failures and co-failures are disclosed in full, and are reported separately from cross-layer binding because they are separate properties. Failed-layer occurrences across the 89 denials are policy 39, evidence 36, state 17, authority 30, measurement 39; state remains the least-exercised layer, and we say so rather than leaving the reader to notice. Every off-diagonal cell of the co-failure matrix is at least 5, and 22 of the 31 possible non-empty failure subsets occur. The cross-layer-binding family contributes nothing to any of these numbers, by construction: its 8 denials have no failed layer at all.

The matrix reports co-failure rather than cross-artifact binding. A populated off-diagonal cell means two layers failed in the same transaction for independent reasons; it is co-occurrence, and it says nothing about whether the two layers were describing the same action. Co-failure coverage and cross-layer binding are two different properties with two different counts: the minimum off-diagonal co-failure cell is 5, and the number of cross-layer denials is 8. Neither number substitutes for the other, and only the second tests the seam.

The typed binding vocabulary is exercised as follows. Because the binding function is pure over the subject records, it is computed for every transaction, and corpus-wide it returns PASS 50, EFFECT_MISMATCH 28, PROFILE_MISMATCH 9, RESOURCE_MISMATCH 3, TIME_MISMATCH 3, and UNDETERMINED 11. Only a subset of those outcomes is *decisive*, that is, reached by the gate ladder because all five layers passed, and only those may be quoted as binding failures: binding.effect 3, binding.resource 2, binding.time 2, binding.measurement_profile 1, binding.undetermined 0. The larger corpus-wide counts are dominated by transactions whose policy is substituted or whose evidence is tampered — those deny earlier, and the binding outcome is never consulted — and all 11 UNDETERMINED transactions carry a MISSING policy, which fires policy.missing first, so that branch is live in the vocabulary but never decisive here. The shipped summary keeps the two counts in separate fields so that the larger cannot be quoted as if it meant the smaller.

Coverage concentration is reported rather than hidden: one authority case (the exact two-hop positive path) backs 59 of the 104 transactions, and all 16 delegation cases appear. Six measurement fixtures appear: the four frozen SAFE fixtures, including the unsupported fixture, exercised five times, which makes the FAIL_POINT composition branch live, plus the two profile variants this laboratory owns for the binding family, which the unmodified SAFE evaluator passes and which differ from each other only in profile identity. The state layer has three failure outcomes plus a pass, and all four states occur: CONTRAINDICATED 9, STALE 4, and REFERENCE_MISMATCH 4, against 87 PASS. Per-case usage tables are in the result summary.

6.4 Companion evidence-class results

The companion laboratories return the following exact designed-fixture results:

Evidence-class change	Executed result	Interpretation ceiling
native signed authority/effect adapter	8/8 verdict + first gate	real Ed25519 over a Tyche test profile; internal labels, no external-protocol conformance

Evidence-class change	Executed result	Interpretation ceiling
single-fault mutations	38/38 oracle pairs; 570/570 Python/JS typed fields	1 baseline + 37 generated faults; frozen-before-comparison, but internally designed
multi-fault mutations	12/12 oracle pairs; 180/180 Python/JS typed fields	2–5 simultaneous faults; internally designed precedence cases
prospective signed revocation	13/13 oracle pairs; 13/13 exact Python/Node rows	scheduled sequences, not a live concurrent status service
native runtime-attestation gate	7/7 verdict-and-gate cases; four native + two binding negatives rejected	tpm2_checkquote over one frozen swtpm vector; no hardware or live workload
identified vTPM VM companion	four-lane contract 20/20; 104/104 live TPM2 quotes; 64/64 native mutation rejects; no new transient or persistent handle	distinct Hyper-V VM/OS and Microsoft vTPM; no physical-host identity, manufacturer-certified EK, or hardware-root claim
live revocation service	276 traces, 890 persisted signed responses; atomic guard 0/0 safety errors	loopback HTTP and one process; safety result, not performance or multi-host evidence
native state transaction overlay	104/104 source classes and gates in Python and compiled Java; 64/64 mutation rejects and parity; 104/104 exact recompositions; Java 19/19 assertions; 8 distinct RSA/ECC AKs	cross-implementation native TPM2 evidence for the full corpus, but software TPM, shared packet, and one physical host
process-isolated durable revocation	372 cases; 1,478 durable events; 96/96 fault recoveries; 0 duplicates; atomic guard 0/0	separate services/processes with restart and retry, but loopback and centralized SQLite on one host
containerized durable revocation	five containers; 135 decisions, 537 events, 95 effects; 540 signed responses; 4/4 fault recoveries; atomic false allows 0; duplicate decisions/effects 0; verifier 15/15	two effect instances and deterministic proxy, but one physical host and shared SQLite volume
typed EATF transfer	21/21 native-code rows plus 104/104 transaction rows mapped; SQL 125/125; 46/46 wrong-class and 46/46 omission detections; sealed blind 46-task semantic kit	analytic author-defined crosswalk; semantic kit undispatched; establishes no interoperability
standalone finite model check	both Boolean outputs; 16/16 binding states; 6/6 one-read and 4/4 two-read schedules; zero atomic-oracle mismatches	implementation-diverse exhaustive check of declared finite models only
isolated container re-execution	PASS for the 104-case corpus and companion evidence upgrades through scheduled revocation	different userland and interpreter; same host, kernel, CPU, and compiled NumPy/OpenSSL wheels

Evidence-class change	Executed result	Interpretation ceiling
optional external-label extension	not executed and not counted	neutral 104-case packet and blind analysis pipeline prepared; no claim of human agreement is made

The revocation strict profile has no false allow against its 13 author-written expectations. The appraisal-only ablation false-allows seven cases: revocation between appraisal and commit, revocation at commit, stale commit status, lower-sequence replay, unavailable commit status, credential expiry at commit, and a corrupted commit-status signature. Commit fail-open false-allows the unavailable-status case; timestamp-only false-allows the lower-sequence replay. These values are intentionally case identifiers and coverage counts, not estimates of how frequently a deployed revocation service races.

The native runtime-attestation gate allows the unmodified vector and matches all seven hash-pinned author-written verdict-and-gate cases. Two negatives — substituted outcome and altered declared PCR — stop at the explicit measurement-binding precheck before the native verifier is called. Fresh-challenge replay, capsule substitution, quote-message tampering, and quote-signature tampering produce non-zero `tpm2_checkquote` returns. This supplies a native verifier call in a bounded companion experiment; it does not upgrade the structural state objects inside the 104 transactions by itself.

The native state overlay supplies that missing corpus-wide upgrade. All 104 quotes pass their cryptographic, PCR, qualifying-data, challenge, and subject checks before policy appraisal. The native results exactly reproduce 87 PASS, 9 CONTRAINDICATED, 4 STALE, and 4 REFERENCE_MISMATCH classes, 104/104. All eight AK public-key fingerprints, all 104 quote messages, and all 104 challenges are distinct. The 64 altered vectors fail at the native evidence gate, and substituting the native state classes into the original four other layer results yields 104/104 exact composed tuples. This closes the structural-main-state and one-vector limitations **within the designed benchmark**. It does not attest a deployed workload: all roots are fresh software TPMs on the same `x86_64` host, the reference policy is author-written, and no manufacturer endorsement or trusted external clock exists.

The independent compiled verifier matches the baseline state class and first gate on 104/104 vectors, rejects all 64/64 independently reconstructed mutations with exact state/gate parity, verifies 104/104 root-to-AK pins, and passes 19/19 predeclared assertions. The implementation source imports only `java.base`; its frozen results are bound to the exact overlay and primary verdict hashes. This removes a single-language/single-parser ceiling, not the shared-fixture, same-operator, software-root, or same-host ceilings.

The live service executes 69 cases per profile, including 64 same-barrier trials. Every one of the 276 result traces verifies its service responses; all 890 exact payload/signature pairs and the raw public key are persisted, and the saved-result checker reverifies them. The atomic guard has 0 false allows and 0 false denies. In this run the double-read, single-read, and TTL-cache profiles have 31, 39, and 31 false allows and no false denies. The forced-expiry case refreshes and denies under `t1_cache` but false-allows under `single_read`, proving that the implementations are behaviorally distinct. The exact three incomplete-profile counts depend on operating-system scheduling in the same-barrier trials and are descriptive observations, not fixed expectations. Their deterministic revoke-after-final-read cases and Proposition 3 establish the existence of the TOCTOU schedule independently of those counts.

The process-isolated extension executes 93 cases per profile, 372 total, through separate status and effect service PIDs and separate contender processes. The durable log of the archived run contains 1,478 events and 301 effects. A case is one scheduled contender sequence against one credential; a recovery is a disconnected or killed-and-restarted commit whose retry returns the original idempotent effect. All 372 case traces verify their signed responses. Across 96 response-loss/retry and kill/restart cases, 96 recover the original idempotent effect and none creates a duplicate. The atomic guard again has 0 false allows and 0 false denies; double-read,

	policy	evidence	state	authority	measurement
policy	39	23	5	5	22
evidence	23	36	5	6	19
state	5	5	17	8	7
authority	5	6	8	30	6
measurement	22	19	7	6	39

Diagonal (bold) = transactions in which the layer failed; off-diagonal = transactions in which both layers failed, over 104 transactions and 89 denies. Every off-diagonal cell is at least 5: the corpus was engineered so that no cell is zero, so cell magnitudes reflect corpus construction, not failure prevalence anywhere. The matrix ranges over the five artefact verifiers only, so the 8 cross-layer denials — denials with an empty failed-layer set — appear in no cell of it. Counts on a designed corpus with author-written labels; they are not rates. State failures are structural-appraisal outcomes, not attestation verdicts of any runtime.

Fig. 3. Pairwise co-failure matrix over the 104 transactions, on a designed corpus engineered so that no off-diagonal cell is zero; cell magnitudes reflect corpus construction, not failure prevalence. Co-occurrence of two layer failures is not a join; the binding stage is not a layer and has no cell here.

single-read, and TTL-cache expose 63, 67, and 66 false allows in that scheduling realization, with no false denies. Those three totals are not rates. The stronger observation is categorical: the atomic invariant and zero-duplicate invariant survive process separation, durable recovery, and ambiguous client outcomes, while every incomplete profile still has a stored counterexample.

The five-container extension records 135 decisions, 537 events, and 95 effects across 69 atomic and 66 prior-read cases. All 540 persisted signed responses reverify. Four injected transport-fault cases recover through the second effect instance; allowed and denied retries replay the original idempotent decision, and no duplicate decision or effect key appears. The atomic profile has zero false allows; the intentionally incomplete prior-read profile exposes 36 in this scheduling realization, including its fixed scheduled witness. Fourteen in-run assertions and 15

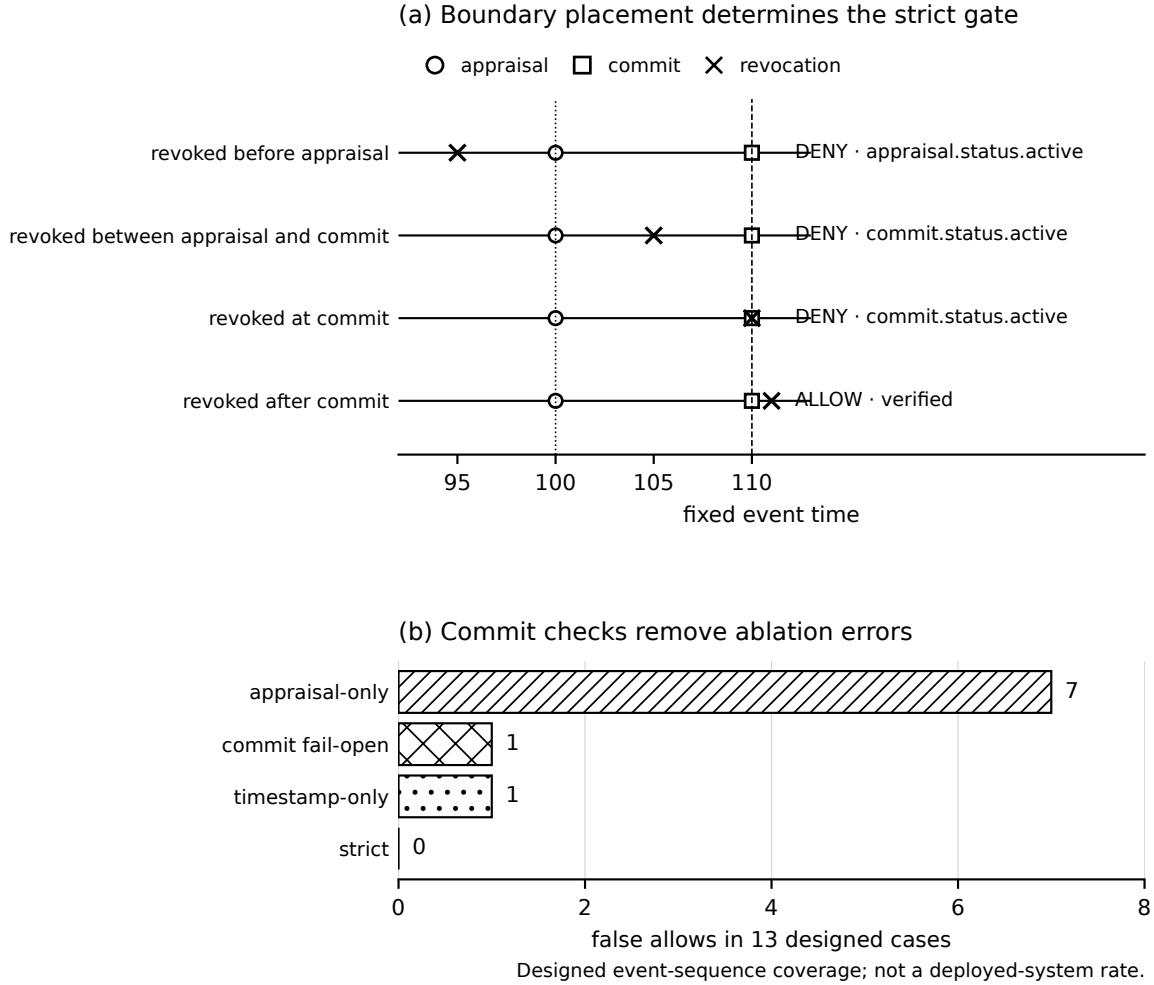
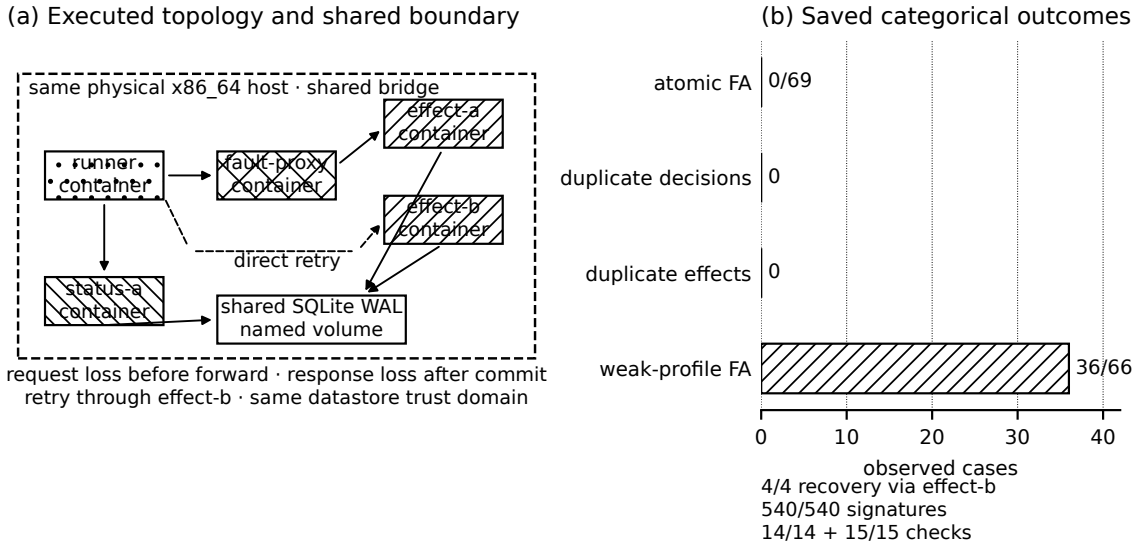


Fig. 4. Four temporal boundary placements in the signed revocation fixtures, with false allows of three incomplete profiles. Revocation at the commit instant is effective under the strict profile; revocation after commit is not retroactive. Coverage counts over 13 designed sequences.

separate saved-result checks pass. These counts add cross-instance and network-fault evidence on one host; they are not distributed-system or rate estimates.

The standalone Java checker accepts all of its finite assertions. For the shared all-pass local vector, each of the two possible Boolean-composer outputs makes at least one error across the matched/mismatched pair. The binding enumeration visits 16/16 agreement states: a Boolean-only rule admits all 15 mismatches; effect/resource binding admits 3; adding profile or time admits 1 in each branch; full effect/resource/time/profile binding admits 0. The revocation enumeration visits all six one-read schedules and all four topologically valid two-read schedules. The atomic oracle has zero mismatches; the incomplete profiles retain the canonical $R < X < C$ and $R1 < R2 < X < C$ false-allow witnesses.



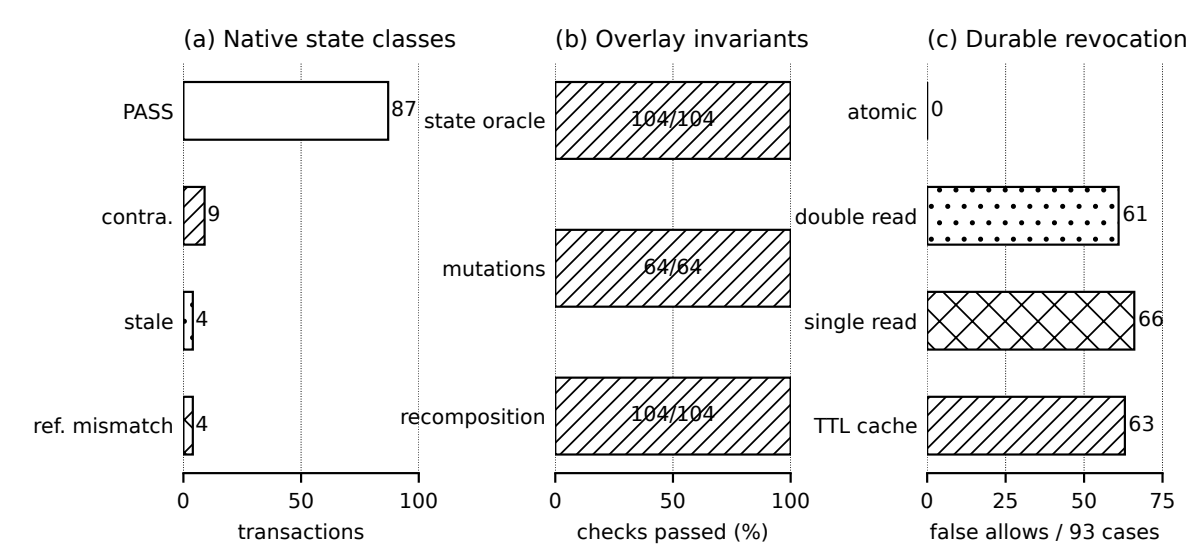
Executed instance separation; one physical host and shared SQLite remain explicit ceilings.

Fig. 5. The executed five-container durable-revocation topology and saved outcomes. Panel (a): runner, deterministic fault proxy, status service, and two effect instances inside one dashed physical-host boundary, with the initial request through the proxy to effect-a, the direct retry to effect-b, and the shared SQLite trust domain. Panel (b): zero atomic false allows, zero duplicate decision/effect keys, 36 designed prior-read counterexamples, 4/4 recoveries through the second instance, 540/540 signature verifications. The physical host and datastore remain deliberately visible ceilings.

The typed-transfer builder maps every native state and refuses partial crosswalks. The EATF result preserves 16 rejection codes across 21/21 TypeScript/Python rows; the transaction result preserves 28 rejection gates across 104/104 rows. Five of eight coarse analytic classes appear in both corpora. If only the Boolean REJECT were retained, the most frequent native code would identify 4/19 EATF rejects (21.1%) and the most frequent gate would identify 14/89 transaction rejects (15.7%); more fundamentally, 498 unordered pairs of distinct native rejection-code types become indistinguishable: 120 within EATF and 378 within the transaction corpus. These are code-type pairs, not observed errors, lost bits, or proof of semantic equivalence. The separately implemented SQLite path reproduces the coarse class for all 125 rows. Changing each of the 46 mapping entries in turn causes at least one mismatch, and removing each entry causes at least one unmapped row. This rules out a vacuous or unused crosswalk entry and adds a separate execution path; it does not make the mapping externally adjudicated.

The blind semantic-validation archive contains exactly 46 distinct native states and four declared annotator files. It contains no author class, expected class, oracle result, or validity flag; the frozen author mapping is represented only by a coordinator-side SHA-256 seal. Two consecutive archive builds are byte-identical and the response validator's synthetic self-test passes. Because no expert has received or returned the packet, these facts establish separation and readiness only, not semantic agreement.

The current offline container run regenerated the 104-transaction corpus with SHA-256 1ba6d40d...a43a8, the native signed fixtures with 09c868dc...583c8, the single-fault Python answers with 3ce82a8c...92e0, the multi-fault Python answers with 8596cb1d...0541, and the revocation corpus with 33625c8d...c563.



96/96 fault recoveries; 0 duplicate effects. Software TPM + same-host loopback/SQLite; coverage, not rates.

Fig. 6. Companion evidence-class upgrades. Panel (a) reports the 104 native state classes. Panel (b) shows exact native-oracle agreement, mutation rejection, and recomposition. Panel (c) contrasts the atomic guard with incomplete revocation profiles over 93 cases each; all 96 injected fault cases recover and no duplicate effect occurs. Counts are designed same-host laboratory coverage, not rates.

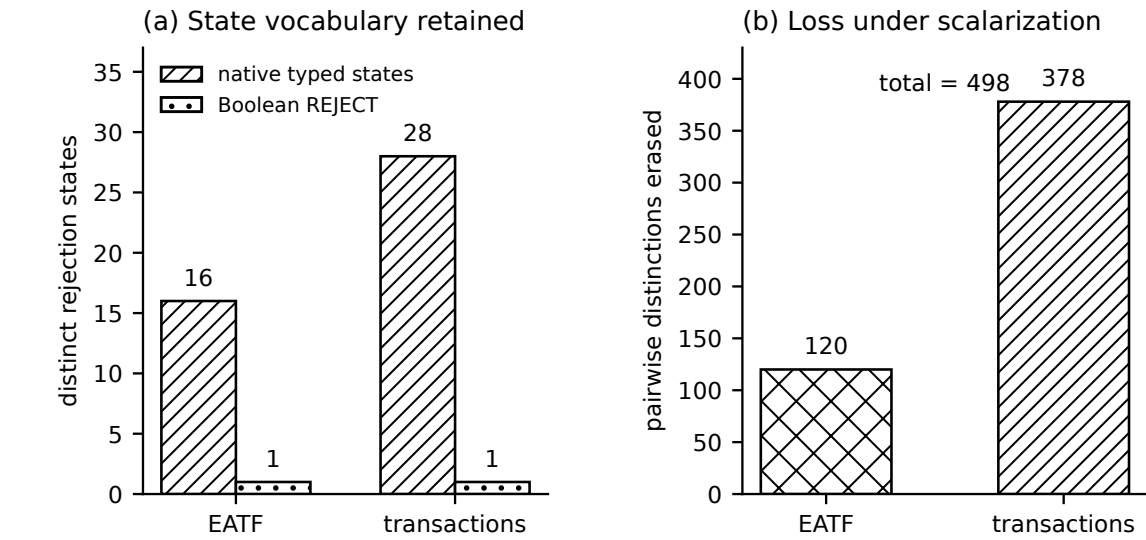


Fig. 7. Typed first-failure states retained in the EATF and transaction corpora, contrasted with one Boolean rejection state. The annotated pair counts are the native distinctions erased by Boolean scalarization, not error rates.

Each full digest matches its host reference. Inside the container only the loopback interface existed, the route table was empty, and an outbound literal-IP TCP connection failed with ENETUNREACH. This closes the current **cross-environment** rerun, not the second-physical-host gate.

One negative provenance result is retained. An independent JavaScript evaluator is frozen against LABELLING-SPEC.md version 1.0, whereas the neutral packet uses version 1.1. The freeze therefore fails the packet's integrity check. Its 90/90 output is a determinacy and implementation-diversity observation about version 1.0, not an independent evaluation of version 1.1 and not an external-human label result. The neutral 104-case packet prepared for future human labelling has not been sent to anyone.

7 Discussion

7.1 Policy is evidence about the appraisal procedure

The policy replay shows why policy should not remain ambient configuration. The same evidence class was allowed or denied depending on whether the verifier checked exact policy identity, version, freshness, digest, and signature. A signed policy was not enough: stale, substituted, and gate-isolated variants remained authentic objects while being inapplicable to the intended decision, and the presence-oriented baselines admitted five and six of them.

Two signed policy fields are operational in the composition: `required_effect` and `required_measurement_profile`. Digest validation alone cannot establish that either agrees with the action actually reported or the measurement profile actually used. The binding stage therefore reads both. `required_effect` is the source of `canonical_action.effect` and is compared against the effect the signed evidence reports and against the terminal granted tools; `required_measurement_profile` is the source of `canonical_action.measurement_profile` and is compared against the profile identifier of the fixture the measurement layer actually appraised. Each comparison has a named gate — `binding.effect` and `binding.measurement_profile` — and each is decisive on the corpus, 3 and 1 transactions respectively. The claim is now supported in the specific sense that removing either field, or writing a different value into it, changes a verdict rather than only a digest.

The honest ceiling is unchanged by this. What the binding stage compares are corpus-supplied strings; matching them establishes that two artifacts agree about a subject, not that either artifact is true. What changed is that a policy field now governs an outcome instead of decorating a hash.

The executed result supports including policy identity, digest, required effect, and required measurement profile in a verifier execution passport; activation evidence and appraisal time are motivated by the same argument but were not exercised by any gate in this corpus. Neither list is established to be complete.

7.2 Measurement support is not a scalar

Two fixtures had point estimates above threshold and still failed the strict measurement rule: one lacked a sufficient lower bound and one carried a different profile identity. The TOPSIS and overlap fixtures expose two more semantics that a scalar value omits: reference-set choice and sample dependence.

The contribution is not that every application must use one confidence level or one reference scheme. The contribution is that those choices must be bound to the evidence identity used to authorize an action.

7.3 Majority voting is the wrong composition

The comparison with the four-of-five majority is best stated as the identity of Section 6.3, in its corrected form: a majority admits every single-fault transaction by definition, and it also admits every transaction in which no artifact verifier failed at all, so on any corpus its false allows are the single-fault denies together with the cross-layer denies. No tuning of the corpus changes that logic. The second term is the more damaging one, and it is invisible to any composition that reduces the layers to a vote: a cross-layer denial has zero failing verifiers,

so a majority over those verifiers has nothing to count. The design is intentionally adversarial to compensation because the predicates answer different questions. A valid policy cannot offset absent authority; an affirming attestation result cannot offset replayed evidence; a high metric cannot offset a substituted effect; and five passing verifiers describing different actions do not add up to one authorized action.

This is analogous to safety interlocks rather than ranking. A scalar may rank alternatives after every critical predicate passes, but it should not purchase permission to ignore a failed critical predicate.

7.4 Exact verifier identity remains a dependency

An internally recorded verifier-provenance differential (public deposit pending) found 11/11 contract matches for the current TypeScript and Python entry points but 8/11, with three false accepts, for an older unversioned demo entry point; a tracked-commit rebuild reproduced the outcome matrix. This article therefore treats exact source and corpus hashes as required artifact inputs and claims nothing further from that record.

7.5 From benchmark seed to evidence-carrying passport

The transaction object suggests a future evidence package:

measurement passport

- + verifier execution passport
- + exact policy snapshot
- + runtime attestation result
- + authority path
- + canonical action record
- + signed reported-effect record (``observed_effect`` in the current schema)
- + typed binding outcome
- + typed composition verdict

Building on the SAFE component and integration families of Kolesnikov and Giudici (Giudici and Kolesnikov 2025, 2026; Kolesnikov 2025), the measurement part can be made explicit as

$$MP = (m, \Sigma, D, \mathcal{P}, \mathcal{A}, \tau, \alpha, t_{\text{eval}}, t_{\text{exp}}, v, \sigma),$$

where m is the component vector, Σ its uncertainty/dependence description, D the data reference, $\mathcal{P}$ the measurement profile, $\mathcal{A}$ the aggregation/reference-set semantics, τ the threshold, α the confidence level, the two times delimit evaluation validity, v identifies the implementation, and σ binds the record. This tuple is our operational evidence proposal, not a tuple attributed to the cited authors. An evidence-carrying decision package then adds the exact policy snapshot, native verifier passports, runtime result, authority path, subject records, decision clock, and typed binding/composition outcomes.

The package should preserve each native verdict and the protected subject it describes. The composed verifier then checks semantic joins rather than relabeling native verifiers as incorrect, and the 8 cross-layer denials are the concrete demonstration of why the subject must travel with the verdict: in each of them every native verdict was correct about its own object, and the package as a whole was still not a basis for the action. In this artifact the two subject records `canonical_action` and `observed_effect` are the minimal form of that requirement.

7.6 Revocation must meet the effect at one boundary

The live race changes the design conclusion. Rechecking status immediately before commit is useful but is not sufficient when the effect occurs in a different operation. The double-read profile still has the schedule `read(active) -> revoke -> effect`. The necessary interface is not simply “fetch fresher status”; it is a guard whose revocation check and effect share one linearization point, or an equivalent transactional protocol that

1458 makes the same invariant externally auditable. This is a systems-design result about the boundary between
 1459 authorization and effect, not a claim that every deployment can place both under one lock.

1460 The decision clock also clarifies what must be carried in evidence. Client timestamps alone cannot adjudicate a
 1461 race with clock skew. A signed, monotonic service sequence or append-only linearization receipt can. A future
 1462 passport should therefore record the revocation object, its sequence, the guarded-commit receipt, and their
 1463 common state-machine identity.

1464 The process-isolated extension narrows an important alternative explanation: the first live result could have
 1465 been dismissed as one in-process lock protecting one in-memory object. The new result keeps the invariant
 1466 when status and effect have different PIDs and listeners, after the effect service loses its response or is killed and
 1467 restarted, because the safety boundary is the durable transaction rather than either process. It still does not show
 1468 a distributed protocol: SQLite is the shared serialization authority, all processes use one host clock and kernel,
 1469 and no partitioned multi-host state was exercised.

1470 7.7 Typed failure is a portable interface, not a universal taxonomy

1471 The EATF transfer supports a modest but useful generalization: different verifiers can preserve their native failure
 1472 vocabularies while exposing the same outer shape — verdict, first decisive state, protected subject, implementation
 1473 identity, and evidence digest. The coarse ontology allows comparison, but its mapping is contestable and must
 1474 remain visibly separate from native semantics.

1475 This is preferable to a forced universal error code. A monitoring system can route `temporal_freshness`
 1476 or `subject_binding` at a coarse level while a domain specialist still sees `TSA_IMPRINT_MISMATCH`, `evidence`,
 1477 `replay`, or `binding.effect`. The 498 collapsed distinctions quantify why retaining only `valid=false` is inade-
 1478 quate for diagnosis, repair, or reviewer audit.

1479 The SQLite mutation audit strengthens a different question from semantic validity. It establishes that every
 1480 mapping entry is live and that two execution paths agree on the declared mapping. Whether `TSA_IMPRINT_`
 1481 `MISMATCH` and `binding.effect` should both be called `subject_binding` remains an interpretive claim until
 1482 independent domain experts or executable external adapters corroborate it.

1483 7.8 Native state evidence can replace a typed fixture without changing composition

1484 The overlay is an evidence-class substitution experiment. The original transaction subjects, other four layer
 1485 results, binding records, and composition rule are held fixed; only the state result is re-derived from a native
 1486 quote and frozen appraisal policy. Exact recomposition 104/104 shows that this substitution is observationally
 1487 conservative at the decision interface while making quote, PCR, challenge, time-window, measurement, and
 1488 transaction bindings falsifiable.

1489 That result does not transform `swtpm` into hardware. Its contribution is the interface pattern: a typed state
 1490 result can be backed by stronger evidence without changing the composition algebra, provided the evidence
 1491 binds the same transaction subject and preserves the typed failure vocabulary. Hardware TPM, manufacturer AK
 1492 provenance, remote verifier sessions, and a deployed workload remain separate experiments.

1493 8 Threats to Validity

1494 8.1 Construct validity

1495 The strict composition reflects the study’s definition of a subject-bound decision over the implemented projection.
 1496 Other systems may use different evaluation orders or distinguish additional layers. The result does not imply
 1497 that the five layers plus the binding stage are a complete ontology, and the binding rules themselves are a design
 1498 choice: four comparisons in a fixed precedence, over subject fields this artifact happens to carry. A deployment
 1499 with a different notion of “same action” would need different rules, and the time comparison here is containment
 1500
 1501
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in a window rather than equality to one timestamp. The main corpus's three decision-relevant constants are unrelated, so it does not instantiate the full decision clock. The live revocation companion instead uses server-side linearization indices for its critical order (Sections 4.2 and 5.6).

The policy corpus uses a test Ed25519 key and an experimental policy profile. The authority objects are not production WAVE, JEDI, OAuth, or MachineMandate credentials, and the authority layer's cryptographic flags are corpus-supplied **in the 104-transaction main corpus**. The companion native adapter and revocation laboratory verify real Ed25519 signatures but use Tyche experimental credential/status profiles and deterministic test keys; they do not retroactively make the main corpus cryptographic or establish conformance with an external protocol. The frozen source corpus still encodes its author-written state oracle in structural attestation-result objects. The native overlay replaces the *evaluation* of that predicate with 104 quote/PCR/challenge/reference-policy appraisals and then recomposes, but it does not make those author-written reference policies external ground truth or attest a deployed runtime. Its eight roots are software TPMs created on one physical host; no hardware root, manufacturer AK identity, protected trusted time, or remote workload exists. The measurement data are synthetic boundary fixtures. The binding-stage ceiling of Section 2 applies unchanged: a cross-layer denial on this corpus demonstrates that the seam is representable and localizable, not that any deployed system exhibits it.

8.2 Internal validity

Expected labels and evaluators originate from the same research programme, and at the composition level the labels are generated by the same shared composition-rule module the verifier imports; that agreement is analytic. Per-layer agreement is the non-analytic execution check; subject-record agreement is explicitly a same-function plumbing check. The same split applies, in a weaker form, to the policy laboratory: there the expected verdict and expected rejecting gate are machine-generated by a precedence ladder that the runner re-implements, so agreement on those two fields is agreement between two transcriptions of one function (Section 6.1).

Pure-Python/NumPy agreement and the relational re-composition paths reduce implementation-path risk but do not create external label independence; only the delegation-path oracle re-derives verdicts from raw case objects, and even it shares case expansion and expected labels with the Python entry point. Three of its four gate-isolation cases exercise a branch by flipping the corpus-supplied Boolean that branch reads, which is branch coverage in two implementations at once and not verification of the property the branch names (Section 5.3).

The recorded Veraison freshness run motivating Section 5.1 and the verifier-provenance differential of Section 7.4 are internal companion artifacts of this research programme; public deposits are pending, and no load-bearing claim depends on either record.

The mutation and revocation expectations are also author-written. A pre-oracle freeze prevents moving evaluator outputs after the mutation oracle is opened, but does not make that oracle independent or correct. The revocation implementations share the same frozen packet and internal expected labels. The independent-JavaScript freeze against the external-label packet targets labelling specification 1.0, whereas the analysis uses specification 1.1; it therefore does not validate the active labelling specification.

Both live services derive their oracle from the state stored by the same linearization mechanism that records the effect. This avoids client-clock guessing but is not an independent implementation of the service semantics. The durable extension narrows the one-process explanation, not the centralized-oracle explanation: both services share one SQLite database. The exact false-allow totals in same-release trials vary with scheduler order; only the zero-error atomic invariant, zero-duplicate invariant, fault recoveries, and deterministic counterexample placements are acceptance conditions. The EATF crosswalk is author-written. Completeness, SQL agreement, and mutation sensitivity are machine-checked, but the choice of eight coarse classes is an interpretive analysis rather than ground truth.

The composed verifier imports source evaluator code. The corpus hash closure prevents silent source-corpus substitution within the run. A companion container run (Section 9) reproduced every reported experimental quantity under a different Linux distribution, CPython patch level, C library, and SQLite engine, but that container shared the host kernel, the host CPU, and the host’s compiled NumPy and OpenSSL binaries. The container run includes the 104-transaction binding corpus, native signed adapter, mutation corpora, and revocation corpus; its cross-environment claim remains a same-host claim (Section 9).

Two subsequent hosted-runner repetitions verify the sealed capsule, apply the same hash-recorded overlay, and execute four declared lanes on separate GitHub-hosted job VMs reporting x86_64 and ARM64, with exact architecture-specific dependency resolution. They satisfy the frozen semantic contract on both architectures and remove the cross-architecture gap for policy replay, composed-corpus evaluation, typed transfer, and process-isolated durable revocation. They do not execute the native-TPM overlay, one-process live service, Java/model-check, or five-container labs on ARM64; identify the underlying physical hosts; use a hardware TPM; or supply outside-operator independence. Local hash-checked runs also cover the native-TPM, live-service, neutral-packet, and typed-transfer companions. The source capsule vendors the SAFE and boundary-seal trees plus an offline amd64 wheelhouse. A hash-bound JSON result and full log record a successful clean temporary extraction on the same x86_64 host. The hosted workflow verifies that exact 210-member source capsule, applies the declared overlay, installs the same pinned dependencies natively for each architecture, and completes its four-lane contract twice on both reported architectures.

8.3 External validity

The 104 deterministic transactions do not sample deployed agent populations. False-allow counts compare profiles on this corpus, are partly determined by corpus composition (Section 6.3), and must not be interpreted as real-world rates. The count of 8 cross-layer denials is likewise a property of the corpus we designed, not a frequency: it says that at least 8 distinct subject-disagreement shapes are representable and were exercised, and it says nothing about how often such disagreements arise anywhere.

The recorded A2A evidence adapter in the authority lineage is hybrid; the experimental authority overlay remains synthetic. Separate local status and effect services, concurrent revocation, response loss, and effect-service restart were executed; a five-container extension adds two effect instances, an internal bridge, and deterministic request/response loss, but shares one physical host and SQLite volume. No public or remote service, payment, robot, institutional workflow, distributed clock, network partition, or multi-host revocation event was exercised, and no latency, throughput, or resource-cost characterization is reported for the live services: the study is correctness-scoped. The neutral external-label packet covers all 104 cases, its blind analysis pipeline is executable, and it remains undispatched.

8.4 Statistical validity

Bootstrap and TOPSIS cases were constructed to expose boundary behavior. They show existence, not prevalence. The bootstrap uses the plain percentile method with n as small as 18 and no BCa correction; the companion seed/replicate sweep (Section 5.2) shows the reported outcomes are not seed or replicate-count artifacts, but the boundary decision could still move under a different estimator family. Across NumPy 2.4.4, 2.4.6, 2.5.0, and 2.5.1 on one processor the lower bounds did not move at all (Section 9), but that probe holds the CPU and its instruction dispatch fixed, so cross-build reproducibility of that third decimal is not established in general. The overlap fixture threshold was declared in the frozen corpus as a designed counterexample. Larger studies must preregister realistic data-generating processes, thresholds, and consequences of false allow and false deny.

The same-barrier and same-release live-race totals are scheduling realizations, not binomial estimates of a stable event probability. No confidence interval is reported for them because the trials are not independent draws

from a defined deployment distribution. Their purpose is to exercise both linearization orders and preserve exact event traces.

9 Reproducibility and Artifact Statement

The laboratory entry points are:

```

labs/policy-version-evidence-replay/run.sh
labs/composed-transaction-corpus/run.sh
labs/formal-composition-modelcheck/run.sh
labs/protocol-valid-unauthorized/verify_corpus.py
labs/protocol-valid-unauthorized/verify_delegation_paths.py
labs/protocol-valid-unauthorized/run_sql_oracle.py
labs/prospective-revocation-races/run.sh
labs/native-runtime-attestation/verify_runtime_attestation.py
labs/native-state-transaction-overlay/run.sh
labs/live-revocation-service/run_live_races.py
labs/distributed-revocation-service/run.sh
labs/containerized-durable-revocation/run.sh
labs/cross-ecosystem-typed-transfer/build_transfer.py
labs/cross-ecosystem-typed-transfer/verify_crosswalk_sql.py
external-label-packet-neutral-104/build_neutral_packet.py
external-label-packet-neutral-104/analyze_responses.py --self-test
companions/native-signed-authority-adapters/verify_fixtures.py
companions/thesis-v4-mutation-tests/evaluate_mutation_corpus.py
companions/thesis-v4-multifault-tests/freeze_and_compare.py
repro/verify_portable_source_capsule.py
repro/run_clean_capsule_replay.py

```

The composed laboratory’s entry point runs, in order, the state-adapter self-test, the corpus build, the cross-layer binding acceptance test, the composed verifier, and the relational oracle, then verifies both checksum manifests. The acceptance test is a gate: it exits non-zero if the Section 1 counterexample is allowed, if any of its five component verifiers fails on that transaction, if the binding result or gate is not EFFECT_MISMATCH / binding.effect, or if the matched lookalike is not allowed at gate verified. The native signed authority adapter, both mutation laboratories, and their input evaluator are committed under companions/; their packet and result digests (09c868dc. . . , af226bac. . . , 3ce82a8c. . . , 5233577a. . . , 8596cb1d. . .) re-derive from the committed trees.

Each deterministic entry point was executed twice end to end against the corpus reported here: every regenerated result file and the complete standard output were byte-identical between runs, and all checksum manifests verified from their owning directories, including the pinned hashes of the frozen SAFE sources. The generated native overlay and both live-race results are intentionally not byte-stable, because fresh attestation-key material, TPM clock fields, operating-system scheduling, and process identity are evidence; their acceptance assertions are typed-class and recomposition parity, mutation rejection, atomic safety, idempotent recovery, signature verification, and exposure of at least one false allow in every incomplete profile. The live entry points bind only to 127.0.0.1 and perform no external request.

A companion container reproduction re-executed the entry points offline from a digest-pinned base image with no network namespace and a read-only root filesystem; it reproduced the composition counts (104 transactions, 15 allows, 89 denies, 8 cross-layer denials), signed adapter 8/8, single-fault 38/38, multi-fault 12/12, and revocation

13/13, with all six regenerated artifact hashes equal to the host references. The container changes Linux userland, CPython patch level, glibc, and SQLite version while sharing the host kernel, CPU, and compiled NumPy/OpenSSL wheels, so it is cross-environment evidence only.

A deterministic 22.1 MB portable source capsule (210 members, embedded manifest, relative-path safety and payload digests verified) vendors the laboratories, their frozen sibling inputs, reproduction scripts, and an offline CPython 3.12 wheelhouse. A declared release overlay — which repairs execute bits on three archived run scripts and replaces one hard-coded architecture label with `platform.machine()`, changing no fixture, evaluator, expected count, or assertion — ran the verified sealed capsule twice on GitHub-hosted x86_64 and ARM64 job VMs. Both repetitions verified 210/210 capsule members, satisfied 20/20 architecture-neutral semantic assertions per architecture, and agreed on all 11/11 cross-architecture comparisons, with identical result counts: 16 policy vectors, 104 composed transactions, eight binding denials, 21 EATF rows, 104 transfer rows, 372 revocation cases, and 96 fault cases. The runs install identical dependency versions natively per architecture. A preserved hosted field reading same physical x86_64/aarch64 host is a legacy topology label from the runner, not physical-host attestation: one job VM per execution, underlying physical identity unknown. The hosted anchors are source commits c85bc3aa. . . (preferred) and 428ce24b. . . (repeat) with their complete downloaded-evidence manifests pinned in the artifact ledger. A direct public-tree matrix run over source commit d605e8d. . . returned 20/20 on the x86_64 job VM, 20/20 on the ARM64 job VM, and 8/8 in the cross-architecture comparison, and a further matrix run at source commit 62d5df8. . . reproduced the same contract; the four lanes' code and fixtures are unchanged between those commits and the present manuscript source.

A complementary execution domain is an identified Ubuntu 24.04 x86_64 Hyper-V VM/OS instance exposing a Microsoft vTPM, recorded in the artifact under the host label zeus2 and distinguished by hashed machine and boot identifiers, DMI strings, and TPM manufacturer properties (MSFT). The identified VM verifies the deterministic public source archive and then satisfies the same four-lane contract (16 policy vectors, 104 composed transactions, eight binding denials, 21 EATF rows, 104 transfer rows, 372 revocation cases, 96 fault cases, 20/20 semantic assertions). Its non-destructive TPM companion first passes 11/11 static/input checks and 17/17 negative safety checks; the vTPM rejects `tpm2_createak` below a generic transient endorsement primary with `TPM_RC_POLICY`, and a declared compatibility overlay instead creates an owner-hierarchy transient primary and an ordinary transient RSA signing key whose encrypted private blob never leaves the temporary private directory. The final run verifies 104/104 quotes with `tpm2_checkquote`, one unique random challenge and qualifying-data digest per transaction, and rejects all 64 predeclared mutations. The identified VM's eight predeclared mutation classes follow the hardware-companion plan (quote message, signature, PCR blob, challenge, transaction substitution, source-corpus hash, capsule hash, and qualifying-data representation); this plan shares six classes with the native-overlay plan of Section 5.6 and replaces its measurement and signed-window classes with the capsule and representation classes. A read-only handle census returns zero new transient and zero persistent handles, and the public bundle contains no `.ctx`, `.priv`, or PEM private key. This is live vTPM evidence on an identified VM/OS, not discrete hardware-TPM or physical-host attestation.

The execution domains and their contract counts are summarized below. Cells are deterministic contract checks on designed corpora, not deployed-system rates; the identified-VM row is a VM/OS with a Microsoft vTPM, not evidence of the underlying physical host or a hardware root of trust.

Execution domain	Policy	Corpus	Transfer	Revocation	Quotes	Mutations
Local x86_64, software TPM (8 fresh swtpm roots; one physical host)	16	104	104	372	104/104	64/64
Hosted x86_64 job VM (contract 20/20; physical host unknown)	16	104	104	372	—	—

Execution domain	Policy	Corpus	Transfer	Revocation	Quotes	Mutations
Hosted ARM64 job VM (contract 20/20; physical host unknown)	16	104	104	372	—	—
Identified VM (zeus2), Hyper-V x86_64 (contract 20/20; Microsoft vTPM)	16	104	104	372	104/104	64/64

The local software-TPM quote and mutation counts are verified independently by the Python and compiled Java implementations; the identified-VM counts are live Microsoft-vTPM verifications.

A blind-safe public artifact repository records Anton Sokolov as author, ORCID 0000-0003-2452-7096, and Tyche Institute as affiliation: <https://github.com/tyche-institute/subject-bound-evidence-composition-artifact>. Original code and schemas carry Apache-2.0 terms; original prose, figures, synthetic corpora, fixtures, and generated research data carry CC-BY-4.0 terms, with upstream notices preserved. The v0.2.0-rc2 release is built by a deterministic fail-closed sanitizer from an exact allowlist over a clean source commit; two builds over the release-candidate tree and one from a clean clone produced byte-identical archives, and a fresh anonymous HTTPS clone of the public commit 212b2ff5. . . independently verified the root manifest, regenerated all four lanes, and satisfied 20/20 semantic assertions and 4/4 replay-evidence manifest entries. Software Heritage completed a full archival visit of the release. The release anchors the code, corpora, and generated evidence, all unchanged in the range between its source commit and the present manuscript source; the preprint copy bundled inside the release is an earlier draft, so the manuscript of record is the journal submission, and a release recut at the submitted commit is planned as a separate, explicitly authorized step. A DOI is not asserted or required by the scoped reproducibility claim.

The root content anchors are:

composed transaction corpus (104 transactions) SHA-256:

1ba6d40d07e62a862b98ec52ab5c189eb491f19e4e1e69a411fba73bbb9a43a8

shared composition-rule module SHA-256:

ac430575a00e0090434202d41c430e9fb9818924fc435ea766c64759fe93e61d

process-isolated durable revocation archived result SHA-256:

367f957503c3959acea1f1ee2ee9bebf506e2eed9878e4a0313423da47879358

portable second-host source capsule SHA-256:

8c8919f9826381da289b73dfd35721938ad5aafea5cd3687b23187589a2d0386

public release archive SHA-256:

9d3f2e38ad64fea29271c31d63ac1b57a35ce68e92152d36edcbda29ac8a867f

Software Heritage snapshot:

swh:1:snp:f04ffb15417761d510910aa98d75add6fda83599

identified-VM provenance / evidence manifest SHA-256:

8da49f87f6c0c589873d618786bee086f03d8030f3f360301118da07f51f86ed
ce20410880c05f4ef0a5a65c8fed9006e63217d5e482e4d1a8395c9ff8ef7dcd

Submitted manuscript, July 2026.

Every other per-artifact digest quoted in Sections 5 and 6 — source corpora, mutation packets, native overlay and model-check results, hosted evidence manifests, and container logs — is recorded in the repository’s checksum manifests (SHA256SUMS, per-laboratory manifests, and the artifact ledger), which the entry points verify as part of their runs. The shared composition-rule module is hashed because it is the single implementation of the conjunction, the binding stage, and the gate ladder imported by both the corpus builder and the verifier: its hash is what makes the analytic character of composition-level agreement checkable rather than merely asserted.

The sanitized artifact repository is public. The source/coordinator repository and the blind-study coordination materials (sealed expected labels, annotator archives, participant coordination records) remain private so that the prepared external-labelling and semantic-validation studies stay blind.

10 Conclusion

RQ1 is answered by the eight cross-layer denials and the ablation ladder of Section 6.3; RQ2 by the native state overlay and the atomic status-and-effect boundary of Section 6.4; RQ3 by the typed transfer result of Section 6.4, at the analytic-crosswalk ceiling stated there.

The artifact study operationalizes a narrow proposition: individually valid or plausible artifacts are insufficient when a required global relation is absent from local verdicts. The executable result instantiates that proposition only for effect, resource, report time, and measurement profile; it does not bind the state/workload subject or prove one complete action identity. Five scalar layer results alone cannot represent a transaction in which all five pass but the transaction remains inadmissible under *C*. The binding stage represents this class, and 8 such transactions exist in the frozen corpus, each denied at a named binding gate with every component verifier passing, each paired with a matched all-valid lookalike that is allowed. The opening counterexample — a correctly signed receipt reporting a WRITE effect where the valid delegation path permits only READ — is denied at binding.effect, while its matched lookalike is allowed.

Across the frozen fixtures, exact policy appraisal avoided the five or six false allows created by weaker policy profiles on the 16-vector corpus. Metamorphic tests exposed profile, uncertainty, dependence, and reference-set semantics absent from a point score. On the 104-transaction composed corpus, point-only, majority, and artifact-validity ablations produced 2, 31, and 48 false allows respectively; the majority count is an identity, but with the single-fault denies *and* the cross-layer denials rather than with the single-fault denies alone, and the failure of the narrower identity is itself the result. The subject-aware partial joins leave 3, 2, and 1 false allows when they omit both time and profile, time only, and profile only, respectively. The counts are corpus coverage rather than rates, and the composed verifier’s 104/104 decomposes into an analytic composition-level component, 104/104 per-layer execution agreement, and a 104/104 same-function subject-record plumbing check, all stated in Section 6.3.

Every companion mechanism — native signed credentials, generated single- and multi-fault transactions, scheduled revocation sequences, native quote appraisal with a separately compiled verifier, the finite model check, process-isolated and five-container durable revocation races, and typed transfer with a separate SQL sensitivity audit — is executable from the committed trees. The live result isolates the systems requirement the scheduled fixtures only suggested: revocation status and effect need one linearization boundary, and the recovery result shows why that boundary also needs idempotent effects. The transfer result isolates one diagnostic cost of Booleanization: 498 unordered within-corpus pairs of distinct rejection-code types collapse to the same Boolean value.

The experimental ceiling is explicit rather than a queue of unearned future claims. The verified capsule plus declared overlay completes its four-lane contract twice on GitHub-hosted job VMs reporting x86_64 and ARM64, and once on the identified Hyper-V VM/OS. The identified VM’s Microsoft-vTPM companion adds 104/104 live quote verifications and 64/64 mutation rejections with a clean post-run handle census. These observations support portability across the reported execution domains and a bounded vTPM evidence path. They do not identify an

underlying physical host, attest a hardware root, measure multi-host consensus, or estimate deployed-system rates.

External human labels remain uncollected because the scoped study does not use them as empirical observations: the designed corpus is evaluated against an explicit author-defined reference specification, and the result claimed is cross-implementation agreement plus counterexample coverage. The undispatched neutral packet is an optional future construct-validity extension. Similarly, the crosswalk is claimed only as an executable author-defined typed adapter whose code preservation and mutation sensitivity are measured; no external semantic consensus or interoperability claim is made. These two extensions may broaden external validity, but their absence does not invalidate the reported synthetic-benchmark result.

The manuscript is therefore a submission candidate for the narrow claim actually tested: local Boolean validity cannot substitute for typed composition over the implemented effect, resource, report-time, and measurement-profile coordinates, and revocation status and effect require one commit boundary in the implemented model. It remains a benchmark and systems result, not a claim of production validity.

Declarations

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Generative-AI assistance. The author used Anthropic Claude Code for unit testing, drafting and editing assistance, reference verification, and reproducibility checks. All claims, results, design decisions, and references are the author’s own and verified by the author, who takes full responsibility. Reported per COPE and ICMJE recommendations.

Data and code availability. The sanitized public artifact repository, deterministic release archive, and Software Heritage snapshot are identified in Section 9, together with the root content anchors.

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11 Reproducibility Checklist for JAIR

All articles

- (1) All claims investigated in this work are clearly stated. **Yes.**
- (2) Clear explanations are given how the work reported substantiates the claims. **Yes.**
- (3) Limitations or technical assumptions are stated clearly and explicitly. **Yes.**
- (4) Conceptual outlines and/or pseudo-code descriptions of the AI methods introduced in this work are provided, and important implementation details are discussed. **Yes.**
- (5) Motivation is provided for design choices, including algorithms, implementation choices, parameters, data sets, and experimental protocols beyond metrics. **Yes.**

Articles containing theoretical contributions

Does this paper make theoretical contributions? **Yes.**

- (1) All assumptions and restrictions are stated clearly and formally. **Yes.**
- (2) Complete proofs or arguments are included for all theoretical claims. **Yes.**
- (3) Informal explanations of the theoretical results are provided. **Yes.**
- (4) Appropriate citations are given for theoretical tools and prior results, including linearizability and artifact-centric MAS verification. **Yes.**
- (5) Novel claims are stated formally as a theorem, propositions, or corollaries. **Yes.**
- (6) Notation is motivated and limited to the local-verdict vector, projections, binding predicate, decision rule, and decision clock. **Yes.**

Articles reporting on computational experiments

Does this paper include computational experiments? **Yes.**

- (1) Required source code is in the licensed, versioned public artifact repository and release. **Yes.**
- (2) The source code has a license allowing free reproducibility use. **Yes—original code and schemas are Apache-2.0.**
- (3) The source code has a license allowing free research use. **Yes—original code and schemas are Apache-2.0.**
- (4) Raw, unaggregated synthetic fixtures and generated evidence are in the public artifact release. **Yes.**
- (5) Unaggregated data have a license allowing free reproducibility use. **Yes—original synthetic corpora and generated data are CC-BY-4.0, subject to preserved upstream exceptions.**
- (6) Unaggregated data have a license allowing general research use. **Yes—original synthetic corpora and generated data are CC-BY-4.0, subject to preserved upstream exceptions.**
- (7) Random-number generation and seeds are described sufficiently for replication. **Yes.**
- (8) Execution environment and infrastructure are described, including relevant software versions and explicit VM, architecture, software-TPM, and vTPM boundaries. **Yes.**
- (9) Evaluation metrics and their motivation are explained. **Yes.**
- (10) The number of runs used for each reported deterministic result is stated. **Yes.**
- (11) Unsuccessful or unsatisfactory experiments are not silently ignored; the specification-freeze drift and initial failed container assertions are retained. **Yes.**
- (12) Analysis extends beyond one-dimensional performance summaries to uncertainty, metamorphic relations, typed gates, and stratified coverage. **Yes.**
- (13) Algorithm and experimental parameter settings are reported. **Yes.**
- (14) Hyper-parameter optimization is not part of the study. **NA.**

- (15) Statistical hypothesis tests for stochastic noise effects are not the evidential design used for these deterministic constructed fixtures. **NA.**

Articles using data sets

Does this work rely on one or more data sets, including benchmark-generator outputs? **Yes.**

- (1) Newly introduced corpora are included in the versioned public artifact with a research-use licence. **Yes.**
- (2) The new data have a license allowing reproducibility use. **Yes—original data are CC-BY-4.0.**
- (3) The new data have a license allowing general research use. **Yes—original data are CC-BY-4.0.**
- (4) Literature and public-source inputs have appropriate citations. **Yes.**
- (5) Literature-derived public sources are publicly available. **Yes.**
- (6) New designed corpora, their construction, provenance, label status, and relevant statistics are described. **Yes.**
- (7) Preprocessing, augmentation, mutation, joining, and bootstrap procedures are described. **Yes.**

Explanations

This is the review-format submission candidate, not a claim of acceptance or publication. The licensed public artifact is a sanitized, exact-allowlist release: coordinator records, expected labels, and active blind-study material are excluded. The source capsule plus a declared overlay completed its four-lane contract twice on GitHub-hosted job VMs reporting x86_64 and ARM64. An identified Hyper-V VM/OS independently completes that contract and adds a 104-quote/64-mutation Microsoft-vTPM companion. None of these records identifies an underlying physical host or proves a hardware root. External human labelling and semantic-panel studies are optional external-validity extensions; no human-agreement or interoperability claim depends on them.